

Faithful Logic Embeddings in HOL: Deep and Shallow, Propositional and Quantified

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Dagstuhl Seminar 26091, Dagstuhl, 24. February 2026
Revisiting the Foundations of Deduction in a New World

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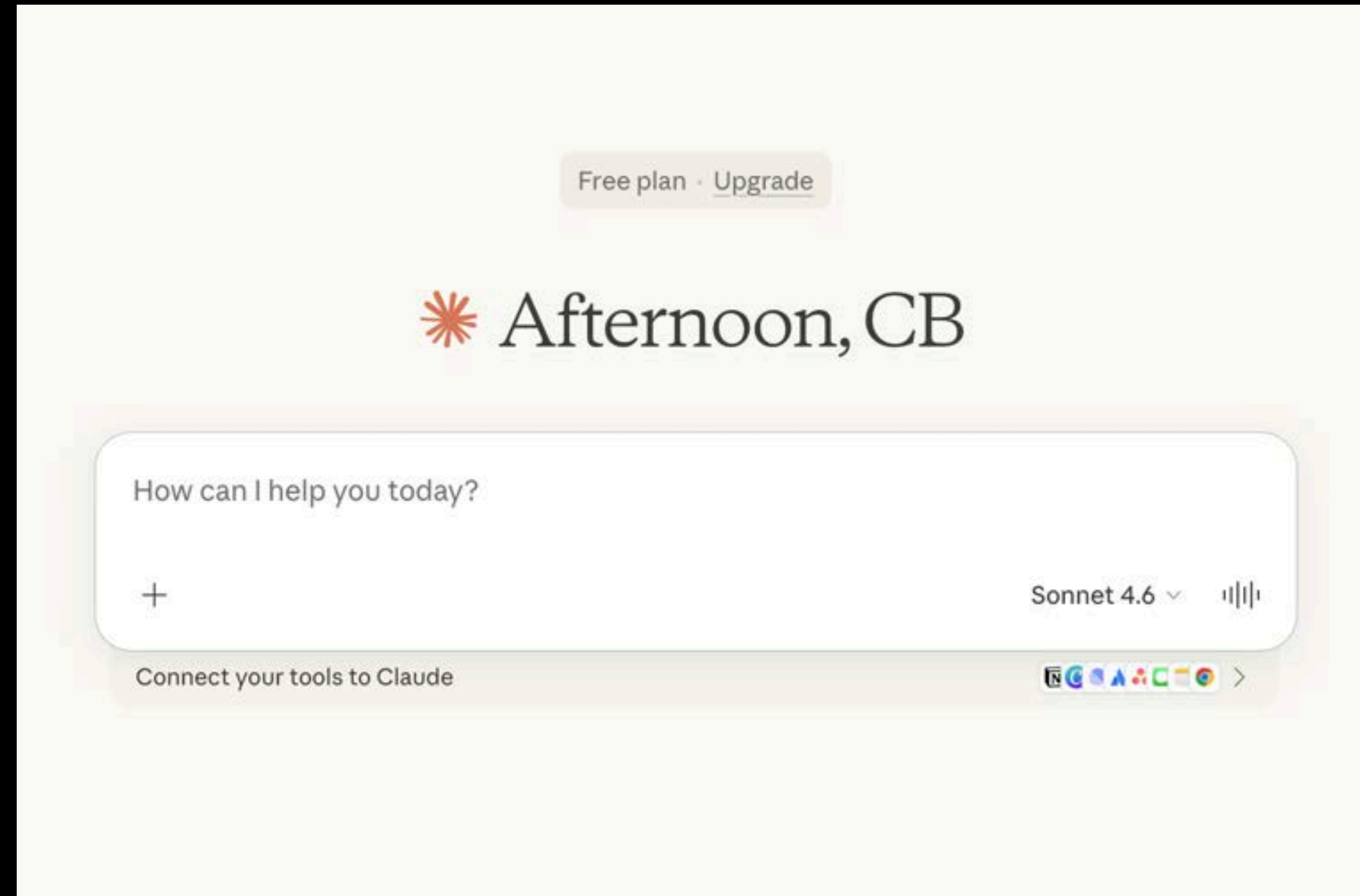


**Significant extension
of CADE 2025 paper**

Dagstuhl Seminar 26091, Dagstuhl, 24. February 2026
Revisiting the Foundations of Deduction in a New World

User interaction & logical pluralism

New world



Even toddlers can use this

Our world

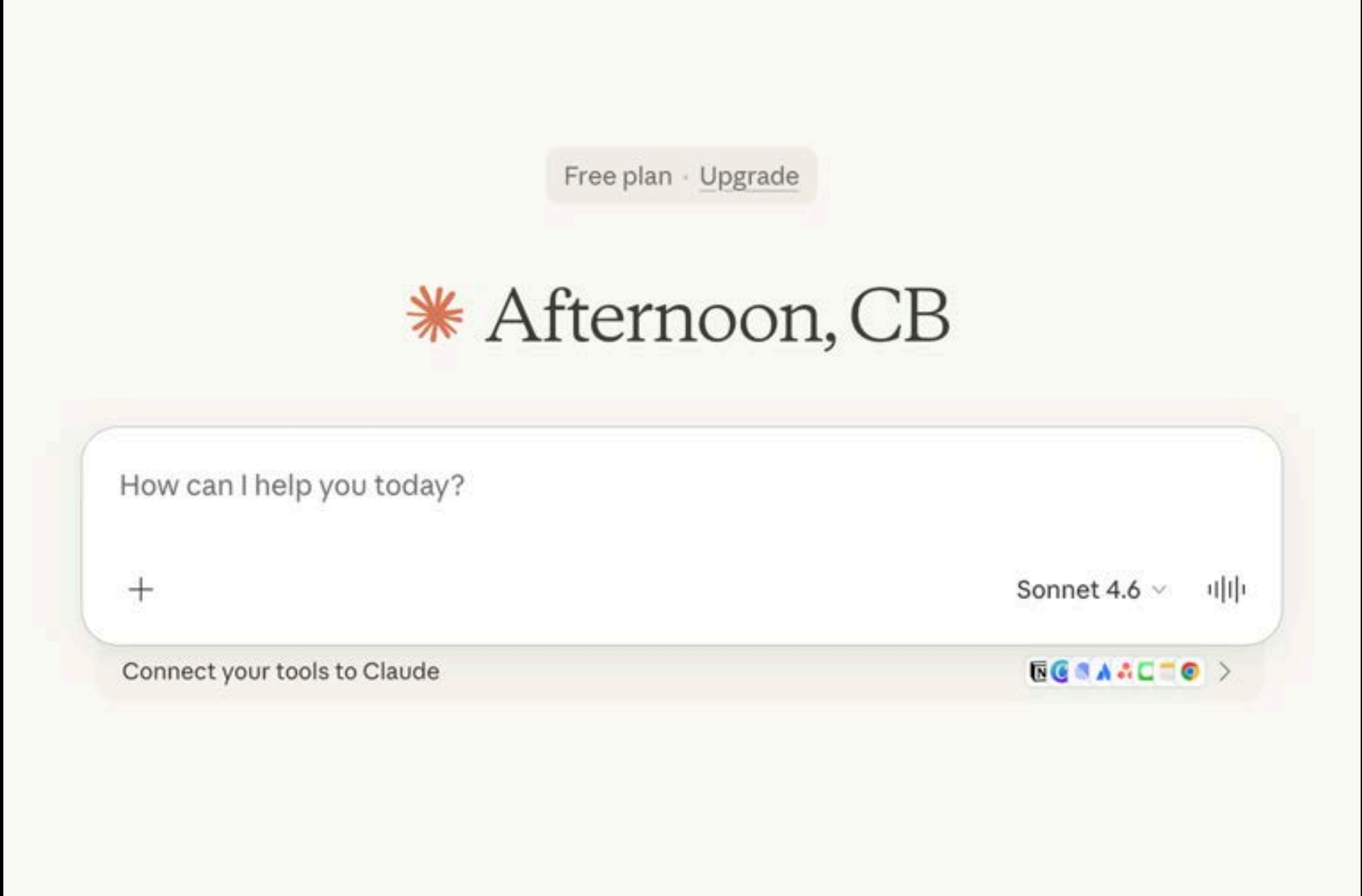


Image generated by DALL-E 3, OpenAI, Feb 2026

Lost users in a zoo of logics and tools

Expressivity matters

New world



No restrictions

JOURNAL ARTICLE

Who finds the short proof?

Christoph Benz Müller, David Fuenmayor, Alexander Steen, Geoff Sutcliffe

Logic Journal of the IGPL, Volume 32, Issue 3, June 2024, Pages 442–464, <https://doi.org/10.1093/jigpal/jzac082>

Published: 25 January 2023 **Article history** ▾

Our world

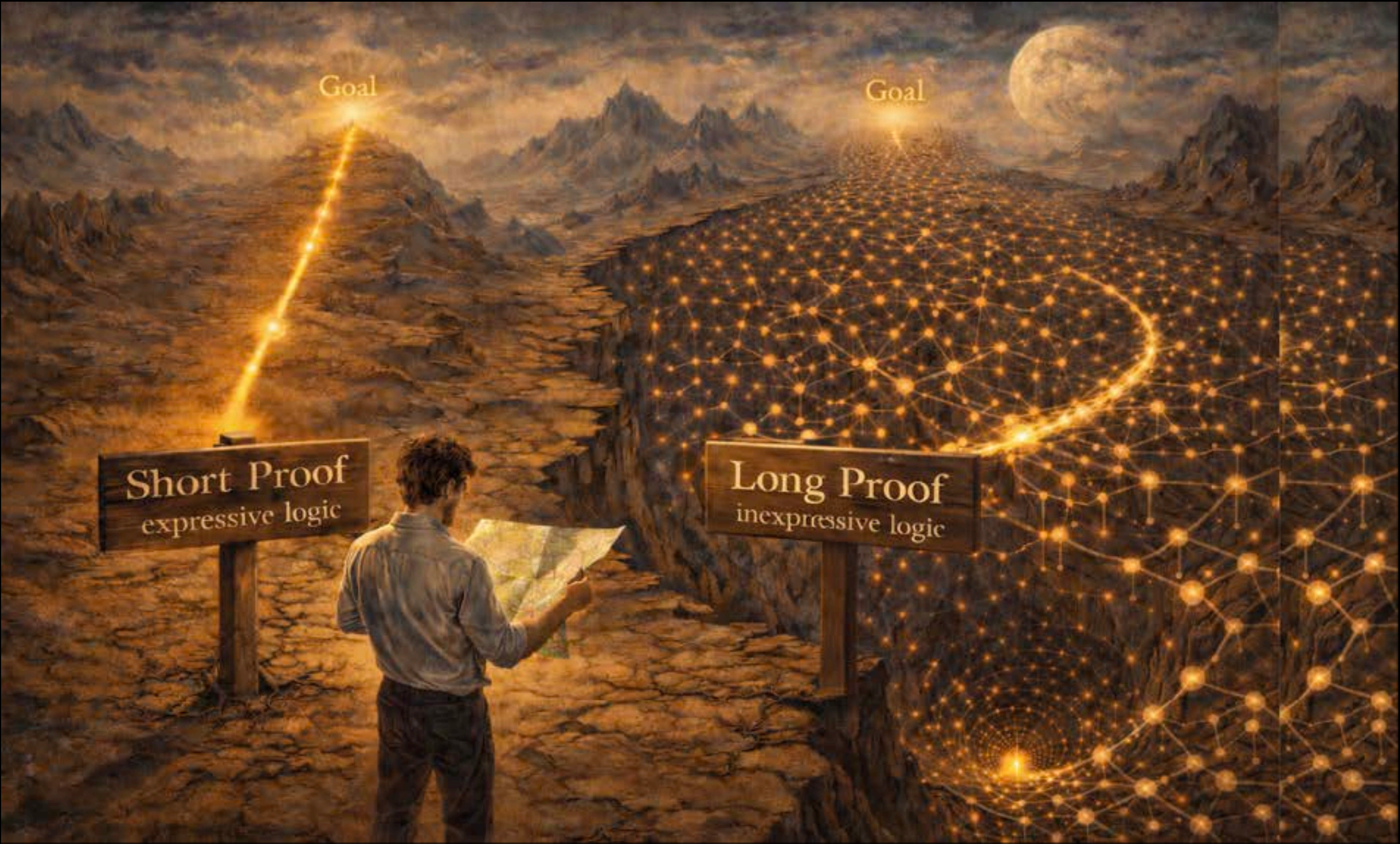
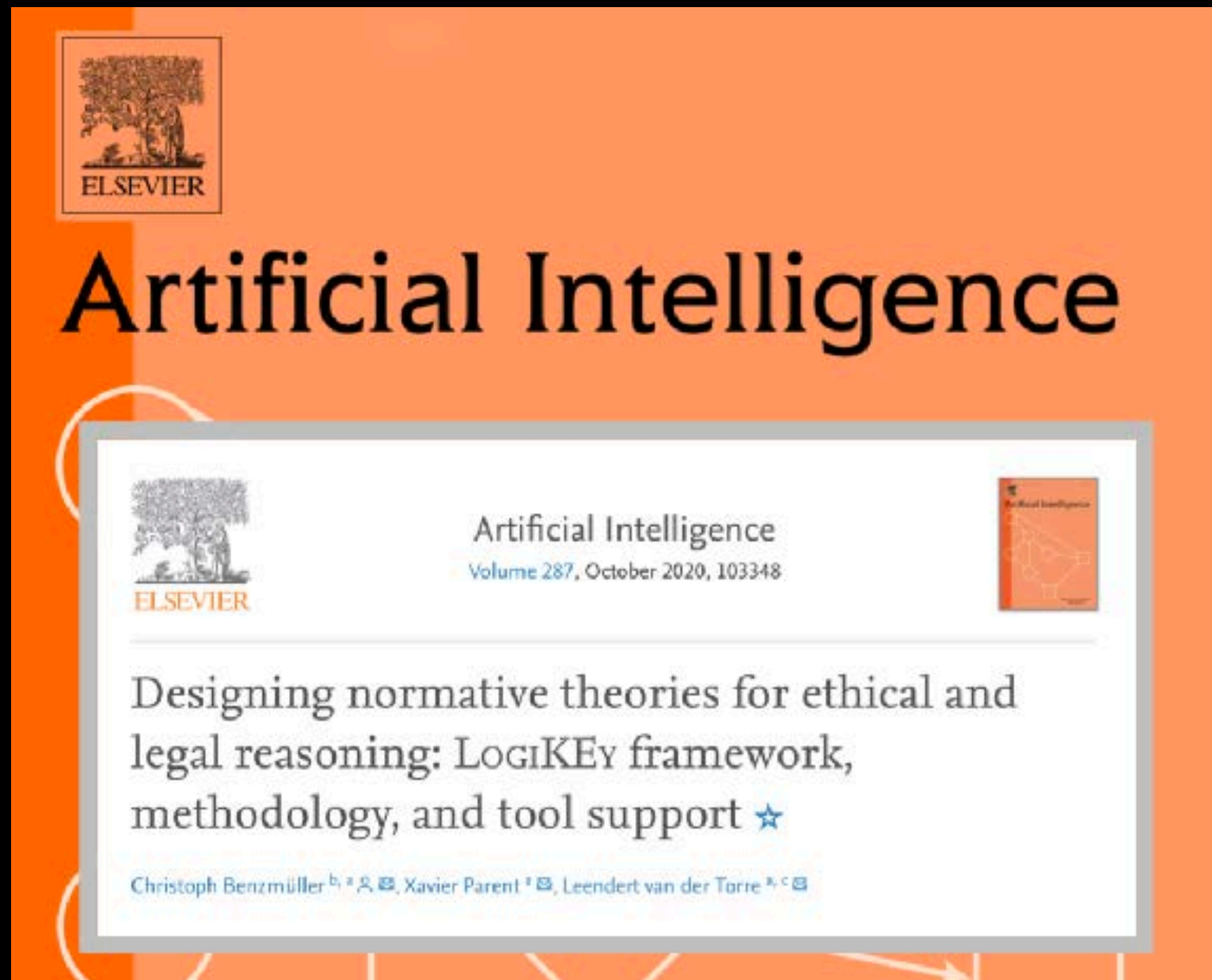


Image generated by DALL-E 3, OpenAI, Feb 2026

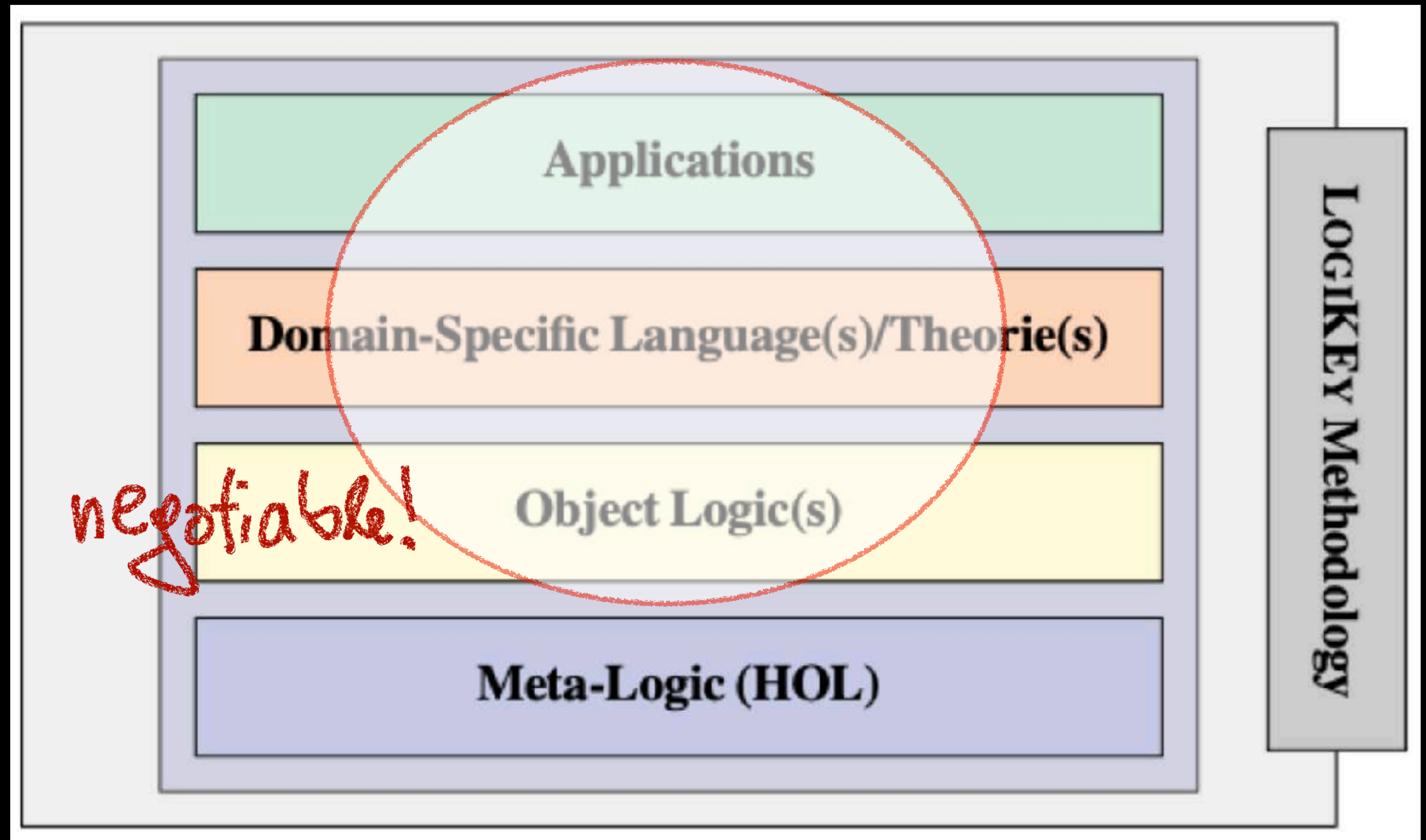
Which logic formalism is best for task?

LogiKEy: Logic-pluralistic KR&R Methodology



2020
term LogiKEy
was introduced

Origins: 2007-2008



LogiKEy: focused so far on shallow embeddings
This talk: integration of deep and shallow embeddings

LogiKEy: Logic-pluralistic KR&R Methodology

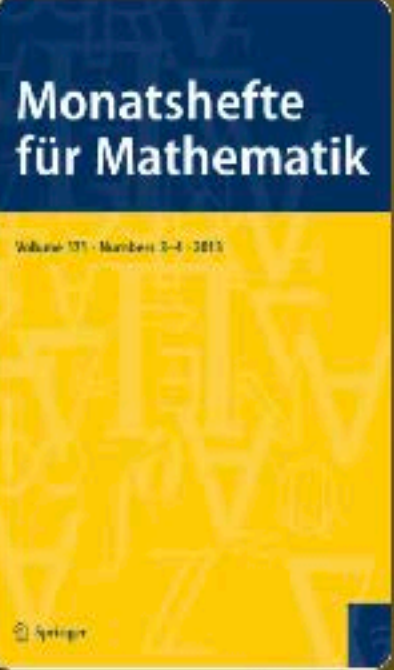
Selection of logic embeddings:

- Multimodal Logic [Andrews Festschrift, 2008]
- Intuitionistic Logic [LJ IGPL, 2010]
- Quantified Multimodal Logic [Log Univers, 2012]
- Quantified Conditional Logic [J Phil Log, 2016]
- Multivalued Logic [LLP, 2016]
- Free Logic [JAR, 2020]
- Dyadic Deontic Logic [AIJ, 2020] [JANCL, 2024]
- Hyper-Intensional HO Modal Logic [RSL, 2020]
- Public Announcement Logic [JLC, 2022]
- Logic of Right to Know [ECAI, 2025]

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Notes on Gödel's and Scott's variants of the ontological argument

[Open access](#) | Published: 21 April 2025
Volume 208, pages 569–611, (2025) [Cite this article](#)



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Christoph Benz Müller  & Dana Scott

2025

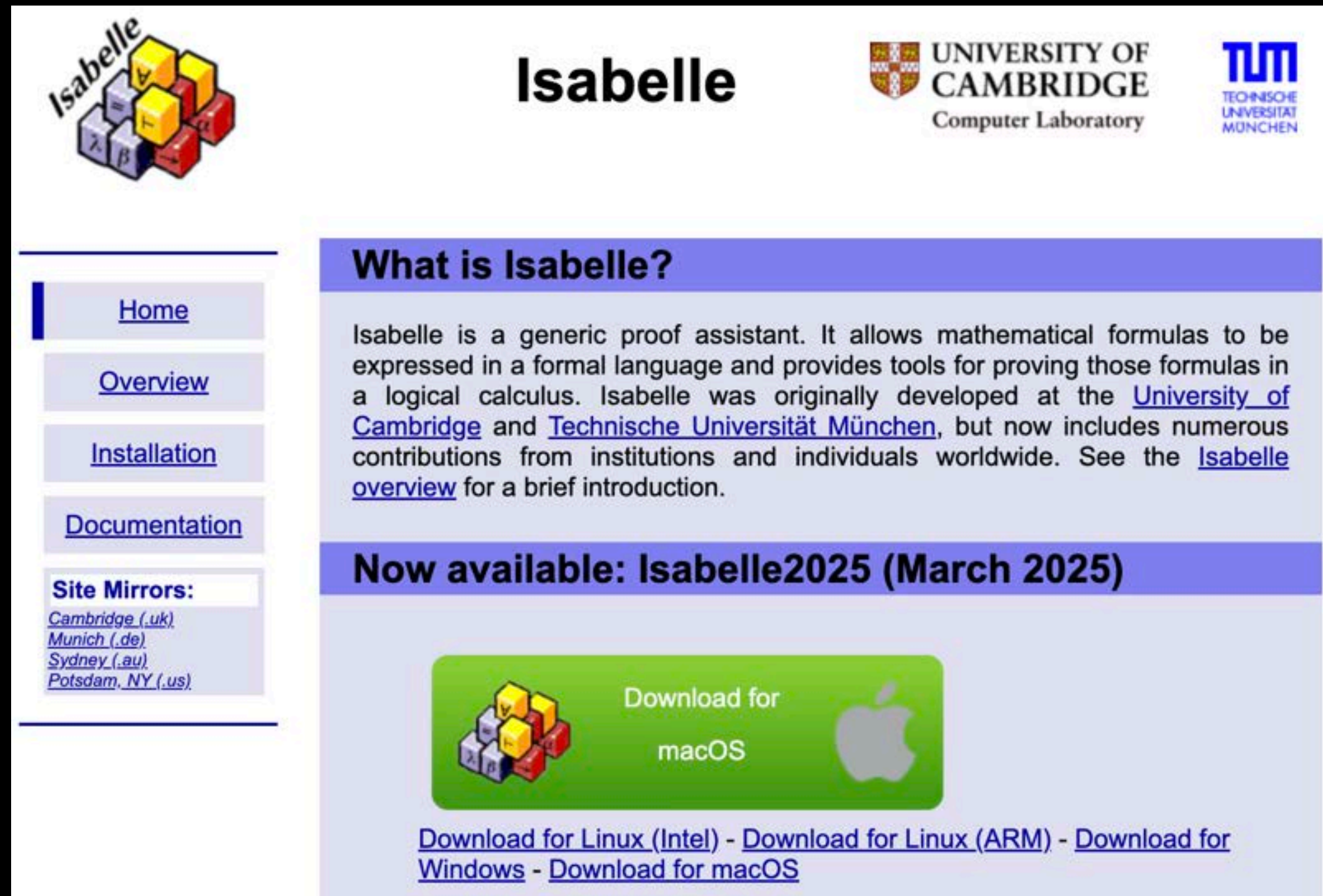
 5405 Accesses  2 Citations  4 Altmetric [Explore all metrics](#) →

Abstract

Notes on Kurt Gödel's modal ontological argument and Dana Scott's variant of it are presented. These remarks, supported by experimental studies with a proof assistant system for classical higher-order logic, implicitly answer some questions the authors have received over the last decade(s). In addition, some new insights resulting from the conducted experiments are reported.

Quantified Multimodal Logics at work: Computational Metaphysics

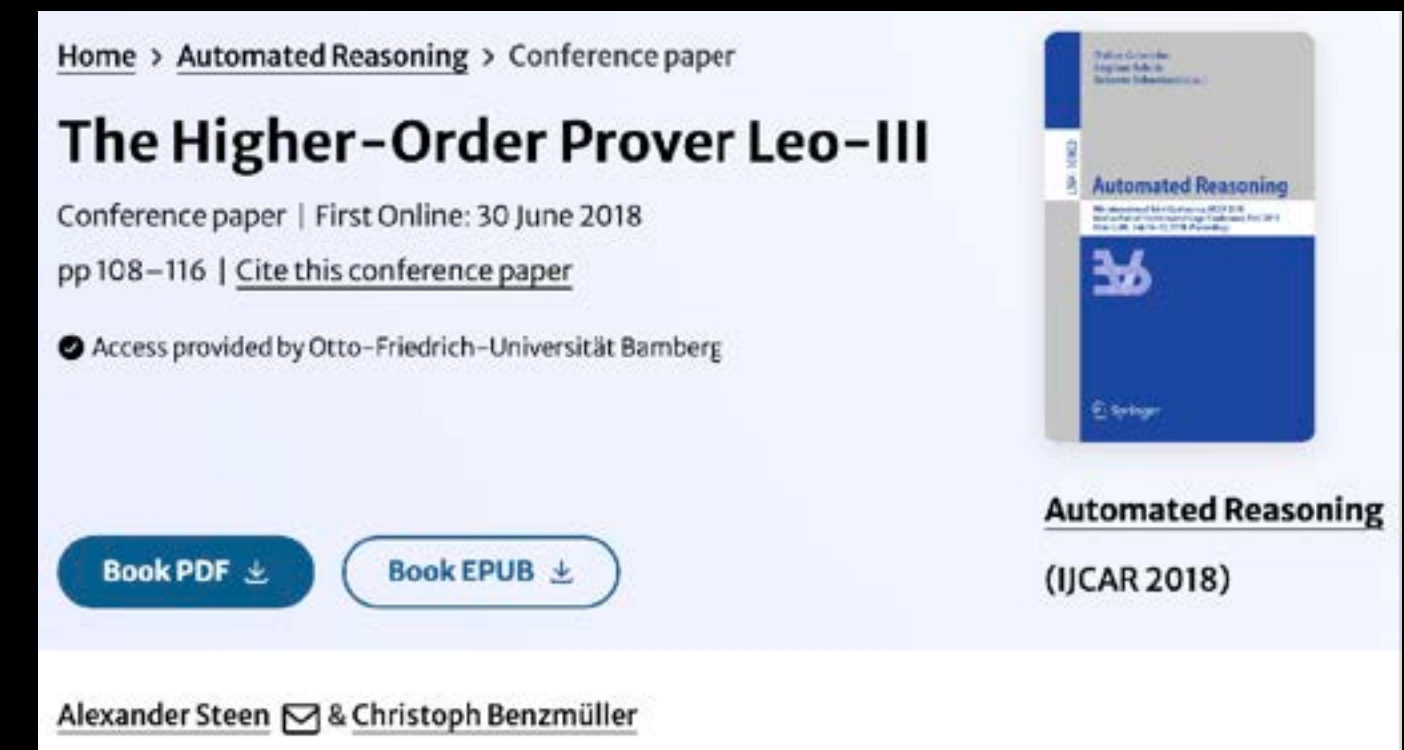
HOL ITP & ATP



The screenshot shows the Isabelle website. At the top, there is a logo for Isabelle (a stack of colored blocks) and the text 'Isabelle'. To the right are logos for the University of Cambridge Computer Laboratory and TUM (Technische Universität München). A left sidebar contains links: Home, Overview, Installation, Documentation, and Site Mirrors (Cambridge (.uk), Munich (.de), Sydney (.au), Potsdam, NY (.us)). The main content area has a heading 'What is Isabelle?' followed by a paragraph describing it as a generic proof assistant. Below this is a blue banner stating 'Now available: Isabelle2025 (March 2025)'. At the bottom, there is a green button for 'Download for macOS' with the Isabelle logo and an Apple logo, and a row of links for 'Download for Linux (Intel)', 'Download for Linux (ARM)', 'Download for Windows', and 'Download for macOS'.



The screenshot shows the article page for 'The Higher-Order Prover LEO-II' in the Journal of Automated Reasoning. The page has a green header with the journal title and a logo. It includes the article title, authors (Christoph Benzmüller, Nik Sultana, Lawrence C. Paulson & Frank TheiB), and publication details (Open access, Published: 22 September 2015, Volume 55, pages 389–404, (2015)). There are buttons for 'Download PDF' and 'Cite this article'. A note states 'You have full access to this open access article'. At the bottom, there are links for 'Aims and scope' and 'Submit manuscript'.



The screenshot shows the conference paper page for 'The Higher-Order Prover Leo-III' in the Automated Reasoning (IJCAR 2018) conference. The page has a light blue header with the conference title and a logo. It includes the paper title, authors (Alexander Steen & Christoph Benzmüller), and publication details (Conference paper, First Online: 30 June 2018, pp 108–116). There are buttons for 'Book PDF' and 'Book EPUB'. A note states 'Access provided by Otto-Friedrich-Universität Bamberg'.

Other provers could be used instead:

- ITPs: Lean, HOL4, Coq, Agda, ...
- ATPs: Vampire, E, Zipperpin, CVC5, ...

Isabelle/HOL: My favorite tool for **logic education** (and choice for this work)

- Proof automation with **Sledgehammer** (FO & HO ATPs, SMT solvers)
- Model- and countermodel-finding with **Nitpick**
- Powerful **User Interface**

Many related papers on **deep and shallow** embeddings in HOL since the early 1990s — see references in CADE 30 paper (and references therein)

My interest here:

- Interlinked logic embeddings in LogiKEy: deep & shallow, prop. **& quantified**
- Reasoning at meta- and object level
- Faithfulness proofs — can they be automated?
- Evaluation — which type of embedding performs best?

Propositional Modal Logic (PML) and **Quantified Boolean Formulas (QBF)** are used in this work only for illustration and reuse in education (courses on universal logical reasoning) — the **QBF part is new !!!**

2 Propositional modal logic (PML)

Syntax

The *syntax* of PML is defined by the grammar $\varphi, \psi := a \mid \neg\varphi \mid \varphi \supset \psi \mid \Box\varphi$, where $a \in \mathcal{S}$ are propositional symbols declared in a (nonempty) signature $\mathcal{S}$. Further logical connectives can be defined as usual, e.g.: $\varphi \vee \psi := \neg\varphi \supset \psi$, $\varphi \wedge \psi := \neg(\varphi \supset \neg\psi)$, $\Diamond\varphi := \neg\Box\neg\varphi$, $\top := p \supset p$ (for some $p \in \mathcal{S}$), and $\perp := \neg\top$.

A *semantics* for PML, called relational semantics or possible-world semantics, determines the truth of a formula φ relative to a so-called possible world w and a relational model. A relational model is a tuple $\langle \mathcal{W}, \mathcal{R}, \mathcal{V} \rangle$, where $\mathcal{W}$ is a set of possible worlds (called the universe), $\mathcal{R}$ is a binary relation on $\mathcal{W}$ (called the accessibility relation), $\mathcal{V}$ is a valuation function that assigns a truth value to each symbol $a \in \mathcal{S}$ in each world $w \in \mathcal{W}$, where $\mathcal{S}$ is the signature of propositional symbols. $\langle \mathcal{W}, \mathcal{R} \rangle$ is called a *frame*. Formally, the truth of a formula φ with respect to a given model $\langle \mathcal{W}, \mathcal{R}, \mathcal{V} \rangle$ and a possible world $w \in \mathcal{W}$ is defined as follows:

$\langle \mathcal{W}, \mathcal{R}, \mathcal{V} \rangle, w \models a$ iff $\mathcal{V}(a)(w)$ is true

$\langle \mathcal{W}, \mathcal{R}, \mathcal{V} \rangle, w \models \neg\varphi$ iff not $\langle \mathcal{W}, \mathcal{R}, \mathcal{V} \rangle, w \models \varphi$

$\langle \mathcal{W}, \mathcal{R}, \mathcal{V} \rangle, w \models \varphi \supset \psi$ iff $\langle \mathcal{W}, \mathcal{R}, \mathcal{V} \rangle, w \models \varphi$ implies $\langle \mathcal{W}, \mathcal{R}, \mathcal{V} \rangle, w \models \psi$

$\langle \mathcal{W}, \mathcal{R}, \mathcal{V} \rangle, w \models \Box\varphi$ iff for all v with $\mathcal{R}wv$ we have $\langle \mathcal{W}, \mathcal{R}, \mathcal{V} \rangle, v \models \varphi$

Semantics

Note that in this recursive definition of truth the only parameter that may be recursively modified is the possible world w in the $\Box$ -clause, which states that a formula φ is “necessary” in a world w iff φ is true at all reachable worlds v from w . Based on the above definitions a notion of validity can be defined: A formula φ is valid (in so called base modal logic **K**) iff $\langle \mathcal{W}, \mathcal{R}, \mathcal{V} \rangle, w \models \varphi$ for all $\mathcal{W}, \mathcal{R}, \mathcal{V}$, and $w \in \mathcal{W}$.

Deep vs. Shallow

(deep)

Gibbons & Wu (2014) Folding domain-specific languages: deep and shallow embeddings (functional pearl). Doi: 10.1145/2628136.2628138

“[...] a **deep embedding** consists of a representation of the abstract syntax as an algebraic datatype, together with some functions that assign semantics to that syntax by traversing the algebraic datatype.

$$\begin{array}{ll} \varphi, \psi := \text{at}(a) \mid \neg\varphi \mid \varphi \supset \psi \mid \Box\varphi & (a \in S) \\ \varphi, \psi := a \mid \neg\varphi \mid \varphi \supset \psi \mid \Box\varphi & (a \in S) \end{array}$$

$\langle W, R, V \rangle, w \models \text{at}(a)$ iff $V(a)(w)$ is true

$\langle W, R, V \rangle, w \models \neg\varphi$ iff not $\langle W, R, V \rangle, w \models \varphi$

$\langle W, R, V \rangle, w \models \varphi \supset \psi$ iff $\langle W, R, V \rangle, w \models \varphi$ implies $\langle W, R, V \rangle, w \models \psi$

$\langle W, R, V \rangle, w \models \Box\varphi$ iff for all $v \in W$ with R_{wv} : $\langle W, R, V \rangle, v \models \varphi$

$\models \varphi$ **iff** $\langle W, R, V \rangle, v \models \varphi$ **for all** W, R, V, v

$$\models \neg\Box a \supset \neg(\Box\neg(a \supset b))$$

Deep vs. Shallow

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“[...] a **deep embedding** consists of a representation of the abstract syntax as an algebraic datatype, together with some functions that assign semantics to that syntax by traversing the algebraic datatype.

```
3 —< Deep embedding (of propositional modal logic in HOL) >
4 datatype PML = AtmD S ("_d") | NotD PML ("¬d") | ImpD PML PML (infixr "⊃d" 93) | BoxD PML ("□d")

11 —< Definition of truth of a formula relative to a model <W,R,V> and a possible world w >
12 primrec RelativeTruthD :: "W ⇒ R ⇒ V ⇒ w ⇒ PML ⇒ bool" ("⟨_,_,_⟩, _ ⊨d _") where
13   "⟨W,R,V⟩, w ⊨d ad      = (V a w)"
14   | "⟨W,R,V⟩, w ⊨d ¬dφ    = (¬ ⟨W,R,V⟩, w ⊨d φ)"
15   | "⟨W,R,V⟩, w ⊨d φ ⊃d ψ = (⟨W,R,V⟩, w ⊨d φ ⟶ ⟨W,R,V⟩, w ⊨d ψ)"
16   | "⟨W,R,V⟩, w ⊨d □dφ    = (∀v:W. R w v ⟶ ⟨W,R,V⟩, v ⊨d φ)"
17 —< Definition of validity >
18 definition ValD ("⊨d _") where "⊨d φ ≡ (∀W R V. ∀w:W. ⟨W,R,V⟩, w ⊨d φ)"
```

Superscript
 d
for "deep"

$$\models^d \neg^d (\Box^d a^d) \supset^d \neg^d (\Box^d (\neg^d (a^d \supset^d b^d)))$$

Deep vs. Shallow

(shallow — maximal)

Gibbons & Wu (2014) Folding domain-specific languages: deep and shallow embeddings (functional pearl). Doi: 10.1145/2628136.2628138

“[...] a **deep embedding** consists of a representation of the abstract syntax as an algebraic datatype, together with some functions that assign semantics to that syntax by traversing the algebraic datatype.

A **shallow embedding** eschews the algebraic datatype, and hence the explicit representation of the abstract syntax of the language; instead, the language is defined directly in terms of its semantics.”

$$\begin{array}{ll} \varphi, \psi := \text{at}(a) \mid \neg\varphi \mid \varphi \supset \psi \mid \Box\varphi & (a \in S) \\ \varphi, \psi := a \mid \neg\varphi \mid \varphi \supset \psi \mid \Box\varphi & (a \in S) \end{array}$$

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$\langle W, R, V \rangle, w \models \varphi \supset \psi$ iff $\langle W, R, V \rangle, w \models \varphi$ implies $\langle W, R, V \rangle, w \models \psi$

$\langle W, R, V \rangle, w \models \Box\varphi$ iff for all $v \in W$ with R_{wv} : $\langle W, R, V \rangle, v \models \varphi$

$\models \varphi$ iff $\langle W, R, V \rangle, v \models \varphi$ for all W, R, V, v

$$\begin{array}{ll} \text{at} & \lambda a . \lambda W R V w . V a w \\ \neg & \lambda \varphi . \lambda W R V w . \neg(\varphi W R V w) \\ \supset & \lambda \varphi . \lambda \psi . \lambda W R V w . (\varphi W R V w) \rightarrow (\psi W R V w) \\ \Box & \lambda \varphi . \lambda W R V w . \forall v:W . R w v \rightarrow (\varphi W R V v) \\ \models & \lambda \varphi . \forall W R V w . (\varphi W R V w) \end{array}$$

$$\models \neg \Box a \supset \neg(\Box \neg(a \supset b))$$

Deep vs. Shallow

(shallow — maximal)

Gibbons & Wu (2014) Folding domain-specific languages: deep and shallow embeddings (functional pearl). Doi: 10.1145/2628136.2628138

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Superscript
 s
for “shallow”

```
3 —<Shallow embedding (of propositional modal logic in HOL)>
4 type_synonym  $\sigma$  = " $\mathcal{W} \Rightarrow \mathcal{R} \Rightarrow \mathcal{V} \Rightarrow w \Rightarrow \text{bool}$ "
5 definition AtmS::" $\mathcal{S} \Rightarrow \sigma$ " ("_ $s$ ") where " $a^s \equiv \lambda W R V w. V a w$ "
6 definition NegS::" $\sigma \Rightarrow \sigma$ " ("¬ $s$ ") where " $\neg^s \varphi \equiv \lambda W R V w. \neg(\varphi W R V w)$ "
7 definition ImpS::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixr " $\supset^s$ " 93) where " $\varphi \supset^s \psi \equiv \lambda W R V w. (\varphi W R V w) \longrightarrow (\psi W R V w)$ "
8 definition BoxS::" $\sigma \Rightarrow \sigma$ " (" $\Box^s$ ") where " $\Box^s \varphi \equiv \lambda W R V w. \forall v:W. R w v \longrightarrow (\varphi W R V v)$ "

15 —<Definition of truth of a formula relative to a model  $\langle W, R, V \rangle$  and a possible world  $w$ >
16 definition RelativeTruthS :: " $\mathcal{W} \Rightarrow \mathcal{R} \Rightarrow \mathcal{V} \Rightarrow w \Rightarrow \sigma \Rightarrow \text{bool}$ " (" $\langle \_, \_, \_ \rangle, \_ \models^s \_$ ") where " $\langle W, R, V \rangle, w \models^s \varphi \equiv \varphi W R V w$ "
17 —<Definition of validity>
18 definition ValS (" $\models^s \_$ ") where " $\models^s \varphi \equiv \forall W R V. \forall w:W. \langle W, R, V \rangle, w \models^s \varphi$ "
```

$\models^s \neg^s (\Box^s a^s) \supset^s \neg^s (\Box^s (\neg^s (a^s \supset^s b^s)))$

Deep vs. Shallow

(shallow — maximal)

Gibbons & Wu (2014) Folding domain-specific languages: deep and shallow embeddings (functional pearl). Doi: 10.1145/2628136.2628138

“[...] a **deep embedding** consists of a representation of the abstract syntax as an algebraic datatype, together with some functions that assign semantics to that syntax by traversing the algebraic datatype.

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$\langle W, R, V \rangle, w \models \varphi \supset \psi$ iff $\langle W, R, V \rangle, w \models \varphi$ implies $\langle W, R, V \rangle, w \models \psi$

$\langle W, R, V \rangle, w \models \Box\varphi$ iff for all $v \in W$ with R_{wv} : $\langle W, R, V \rangle, v \models \varphi$

$\models \varphi$ iff $\langle W, R, V \rangle, v \models \varphi$ for all W, R, V, v

$$\begin{array}{ll} \text{at} & \lambda a . \lambda W R V w . V a w \\ \neg & \lambda \varphi . \lambda W R V w . \neg(\varphi W R V w) \\ \supset & \lambda \varphi . \lambda \psi . \lambda W R V w . (\varphi W R V w) \rightarrow (\psi W R V w) \\ \Box & \lambda \varphi . \lambda W R V w . \forall v:W . R w v \rightarrow (\varphi W R V v) \\ \models & \lambda \varphi . \forall W R V w . (\varphi W R V w) \end{array}$$

$$\models \neg \Box a \supset \neg(\Box \neg(a \supset b))$$

Deep vs. Shallow

(shallow — minimal)

Gibbons & Wu (2014) Folding domain-specific languages: deep and shallow embeddings (functional pearl). Doi: 10.1145/2628136.2628138

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$\langle W, R, V \rangle, w \models \varphi \supset \psi$ iff $\langle W, R, V \rangle, w \models \varphi$ implies $\langle W, R, V \rangle, w \models \psi$

$\langle W, R, V \rangle, w \models \Box\varphi$ iff for all $v \in W$ with $R w v$: $\langle W, R, V \rangle, v \models \varphi$

$\models \varphi$ iff $\langle W, R, V \rangle, v \models \varphi$ for all W, R, V, v

$$\begin{array}{ll} \text{at} & \lambda a . \lambda w . V a w \\ \neg & \lambda \varphi . \lambda w . \neg(\varphi w) \\ \supset & \lambda \varphi . \lambda \psi . \lambda w . (\varphi w) \rightarrow (\psi w) \\ \Box & \lambda \varphi . \lambda w . \forall v . R w v \rightarrow \varphi v \\ \models & \lambda \varphi . \forall w . \varphi w \end{array}$$

$$\models \neg \Box a \supset \neg(\Box \neg(a \supset b))$$

Deep vs. Shallow

(shallow — minimal)

Gibbons & Wu (2014) Folding domain-specific languages: deep and shallow embeddings (functional pearl). Doi: 10.1145/2628136.2628138

“[...] a **deep embedding** consists of a representation of the abstract syntax as an algebraic datatype, together with some functions that assign semantics to that syntax by traversing the algebraic datatype.

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Superscript

m

for “min. shallow”

```
3 —<The accessibility relation R and the valuation function V are introduced as constants at the meta-level HOL>
4 consts R:: $\mathcal{R}$  V:: $\mathcal{V}$ 
5 —<Shallow embedding (of propositional modal logic in HOL)>
6 type_synonym  $\sigma$  = "w $\Rightarrow$ bool"
7 definition AtmM::" $\mathcal{S}\Rightarrow\sigma$ " ("_ $^m$ ") where "a $^m \equiv \lambda w. V\ a\ w$ "
8 definition NegM::" $\sigma\Rightarrow\sigma$ " ("¬ $^m$ ") where "¬ $^m\ \varphi \equiv \lambda w. \neg\varphi\ w$ "
9 definition ImpM::" $\sigma\Rightarrow\sigma\Rightarrow\sigma$ " (infixr " $\supset^m$ " 93) where " $\varphi\ \supset^m\ \psi \equiv \lambda w. \varphi\ w \longrightarrow \psi\ w$ "
10 definition BoxM::" $\sigma\Rightarrow\sigma$ " ("□ $^m$ ") where "□ $^m\ \varphi \equiv \lambda w. \forall v. R\ w\ v \longrightarrow \varphi\ v$ "

17 —<Definition of truth of a formula relative to a model  $\langle \mathcal{W}, R, V \rangle$  and a possible world w>
18 definition RelativeTruthM :: "w $\Rightarrow\sigma\Rightarrow$ bool" ("_  $\models^m$  _") where "w  $\models^m\ \varphi \equiv \varphi\ w$ "
19 —<Definition of validity>
20 definition ValM ("⊨ $^m$  _") where "⊨ $^m\ \varphi \equiv \forall w::w. w \models^m\ \varphi$ "
```

$\models^m \neg^m (\Box^m a^m) \supset^m \neg^m (\Box^m (\neg^m (a^m \supset^m b^m)))$

Example (minimal shallow):

$$\models \neg \Box a \supset \neg(\Box \neg(a \supset b))$$

$$\models^m \neg^m (\Box^m a^m) \supset^m \neg^m (\Box^m (\neg^m (a^m \supset^m b^m)))$$

$$\models \neg \Box a \supset \neg(\Box \neg(a \supset b))$$

$$\neg \downarrow \lambda \varphi . \lambda w . \neg(\varphi w)$$

$$\models (\lambda \varphi . \lambda w . \neg(\varphi w)) \Box a \supset (\lambda \varphi . \lambda w . \neg(\varphi w))(\Box (\lambda \varphi . \lambda w . \neg(\varphi w))(a \supset b))$$

$$\Box \downarrow \lambda \varphi . \lambda w . \forall v . R w v \rightarrow \varphi v$$

$$\models (\lambda \varphi . \lambda w . \neg(\varphi w))((\lambda \varphi . \lambda w . \forall v . R w v \rightarrow \varphi v) a) \supset (\lambda w . \neg((\lambda \varphi . \lambda w . \forall v . R w v \rightarrow \varphi v)(\lambda \varphi . \lambda w . \neg(\varphi w))(a \supset b)) w))$$

$$\supset \downarrow \lambda \varphi . \lambda \psi . \lambda w . (\varphi w) \rightarrow (\psi w)$$

$$\models \dots \dots$$

$$\text{at} \downarrow \lambda a . \lambda w . (V a w)$$

$$\models \dots \dots$$

$$\models \downarrow \lambda \varphi . \forall w . (\varphi w)$$

$$\dots \dots \dots$$

$$\downarrow \beta\text{-reduction}$$

$$\forall w . \neg(\forall v . R w v \rightarrow V a v) \rightarrow \neg(\forall v . R w v \rightarrow \neg(V a v \rightarrow V b v))$$

$$\downarrow \text{simplification}$$

$$\forall w . (\exists v . R w v \wedge \neg V a v) \rightarrow (\exists v . R w v \wedge (V a v \rightarrow V b v))$$

Standard translation of
PML to FOL via unfolding
and β -reduction in HOL

Example (all embeddings):

$$\models \neg \Box a \supset \neg(\Box \neg(a \supset b))$$

shallow minimal ($\cdot^m$)

$$\models^m \neg^m (\Box^m a^m) \supset^m \neg^m (\Box^m (\neg^m (a^m \supset^m b^m)))$$

$$\forall w. (\exists v. R w v \wedge \neg V a v) \longrightarrow (\exists v. R w v \wedge (V a v \longrightarrow V b v))$$

shallow maximal ($\cdot^s$)

$$\models^s \neg^s (\Box^s a^s) \supset^s \neg^s (\Box^s (\neg^s (a^s \supset^s b^s)))$$

$$\begin{array}{l} \forall W R V x. \\ W x \longrightarrow \\ (\exists xa. W xa \wedge R x xa \wedge \neg V a xa) \longrightarrow (\exists xa. W xa \wedge R x xa \wedge (V a xa \longrightarrow V b xa)) \end{array}$$

deep ($\cdot^d$)

$$\models^d \neg^d (\Box^d a^d) \supset^d \neg^d (\Box^d (\neg^d (a^d \supset^d b^d)))$$

$$\begin{array}{l} \forall W R V x. \\ W x \longrightarrow \\ (\exists xa. W xa \wedge R x xa \wedge \neg V a xa) \longrightarrow (\exists xa. W xa \wedge R x xa \wedge (V a xa \longrightarrow V b xa)) \end{array}$$

Example (all embeddings):

~~$$\vdash \neg \Box a \supset \neg (\Box \neg (a \supset b))$$~~

$$\vdash \neg \Box \varphi \supset \neg (\Box \neg (\varphi \supset \psi))$$

shallow minimal ($\cdot^m$)

$$\models^m \neg^m (\Box^m \varphi) \supset^m \neg^m (\Box^m (\neg^m (\varphi \supset^m \psi)))$$

shallow maximal ($\cdot^s$)

$$\models^s \neg^s (\Box^s \varphi) \supset^s \neg^s (\Box^s (\neg^s (\varphi \supset^s \psi)))$$

deep ($\cdot^d$)

$$\models^d \neg^d (\Box^d \varphi) \supset^d \neg^d (\Box^d (\neg^d (\varphi \supset^d \psi)))$$

Schematic formulas φ
lead to more complex
proof problems

$$\forall w. (\exists v. R\ w\ v \wedge \neg \varphi\ v) \longrightarrow (\exists v. R\ w\ v \wedge (\varphi\ v \longrightarrow \psi\ v))$$

$$\begin{aligned} &\forall W\ R\ V\ x. \\ &\quad W\ x \longrightarrow \\ &\quad (\exists xa. W\ xa \wedge R\ x\ xa \wedge \neg \varphi\ W\ R\ V\ xa) \longrightarrow \\ &\quad (\exists xa. W\ xa \wedge R\ x\ xa \wedge (\varphi\ W\ R\ V\ xa \longrightarrow \psi\ W\ R\ V\ xa)) \end{aligned}$$

$$\begin{aligned} &\forall W\ R\ V\ x. \\ &\quad W\ x \longrightarrow \\ &\quad (\exists xa. \\ &\quad \quad W\ xa \wedge \\ &\quad \quad R\ x\ xa \wedge \\ &\quad \quad \neg \text{rec_PML } (\lambda a\ W\ R\ V. V\ a) (\lambda \varphi\ \varphi'\ W\ R\ V\ w. \neg \varphi'\ W\ R\ V\ w) \\ &\quad \quad \quad (\lambda \varphi\ \psi\ \varphi'\ \psi'\ W\ R\ V\ w. \varphi'\ W\ R\ V\ w \longrightarrow \psi'\ W\ R\ V\ w) \\ &\quad \quad \quad (\lambda \varphi\ \varphi'\ W\ R\ V\ w. \forall x. W\ x \longrightarrow R\ w\ x \longrightarrow \varphi'\ W\ R\ V\ x) \varphi\ W\ R\ V\ xa) \longrightarrow \\ &\quad (\exists xa. \\ &\quad \quad W\ xa \wedge \\ &\quad \quad R\ x\ xa \wedge \\ &\quad \quad (\text{rec_PML } (\lambda a\ W\ R\ V. V\ a) (\lambda \varphi\ \varphi'\ W\ R\ V\ w. \neg \varphi'\ W\ R\ V\ w) \\ &\quad \quad \quad (\lambda \varphi\ \psi\ \varphi'\ \psi'\ W\ R\ V\ w. \varphi'\ W\ R\ V\ w \longrightarrow \psi'\ W\ R\ V\ w) \\ &\quad \quad \quad (\lambda \varphi\ \varphi'\ W\ R\ V\ w. \forall x. W\ x \longrightarrow R\ w\ x \longrightarrow \varphi'\ W\ R\ V\ x) \varphi\ W\ R\ V\ xa \longrightarrow \\ &\quad \quad \text{rec_PML } (\lambda a\ W\ R\ V. V\ a) (\lambda \varphi\ \varphi'\ W\ R\ V\ w. \neg \varphi'\ W\ R\ V\ w) \\ &\quad \quad \quad (\lambda \varphi\ \psi\ \varphi'\ \psi'\ W\ R\ V\ w. \varphi'\ W\ R\ V\ w \longrightarrow \psi'\ W\ R\ V\ w) \\ &\quad \quad \quad (\lambda \varphi\ \varphi'\ W\ R\ V\ w. \forall x. W\ x \longrightarrow R\ w\ x \longrightarrow \varphi'\ W\ R\ V\ x) \psi\ W\ R\ V\ xa)) \end{aligned}$$

Interlinked Embeddings

shared preliminaries

—<Type declarations common for both the deep and shallow embedding>
typedec1 w —<Type for possible worlds>
typedec1 $\mathcal{S}$ —<Type for propositional constant symbols>
consts $p::\mathcal{S}$ $q::\mathcal{S}$ —<Some propositional constant symbols>
type_synonym $\mathcal{W}$ = " $w \Rightarrow \text{bool}$ " —<Type for sets of possible worlds>
type_synonym $\mathcal{R}$ = " $w \Rightarrow w \Rightarrow \text{bool}$ " —<Type for accessibility relations>
type_synonym $\mathcal{V}$ = " $\mathcal{S} \Rightarrow w \Rightarrow \text{bool}$ " —<Type for valuation functions>

shallow maximal ($\cdot^s$)

```
1 theory PMLinHOL_shallow imports PMLinHOL_preliminaries (* Christoph Benzmüller, 2025 *)
2 begin
3 —<Shallow embedding (of propositional modal logic in HOL)>
4 type_synonym  $\sigma$  = " $\mathcal{W} \Rightarrow \mathcal{R} \Rightarrow \mathcal{V} \Rightarrow w \Rightarrow \text{bool}$ "
5 definition AtmS::" $\sigma \Rightarrow \sigma$ " ("  $\cdot^s$  ") where " $a^s \equiv \lambda W R V w. V a w$ "
6 definition NegS::" $\sigma \Rightarrow \sigma$ " ("  $\neg^s$  ") where " $\neg^s \varphi \equiv \lambda W R V w. \neg(\varphi W R V w)$ "
7 definition ImpS::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixr " $\supset^s$ " 93) where " $\varphi \supset^s \psi \equiv \lambda W R V w. (\varphi W R V w) \longrightarrow (\psi W R V w)$ "
8 definition BoxS::" $\sigma \Rightarrow \sigma$ " ("  $\Box^s$  ") where " $\Box^s \varphi \equiv \lambda W R V w. \forall v:W. R w v \longrightarrow (\varphi W R V v)$ "
9 —<Further logical connectives as definitions>
10 definition OrS (infixr " $\vee^s$ " 92) where " $\varphi \vee^s \psi \equiv \neg^s \varphi \supset^s \psi$ "
11 definition AndS (infixr " $\wedge^s$ " 95) where " $\varphi \wedge^s \psi \equiv \neg^s (\varphi \supset^s \neg^s \psi)$ "
12 definition DiaS ("  $\Diamond^s$  ") where " $\Diamond^s \varphi \equiv \neg^s (\Box^s (\neg^s \varphi))$ "
13 definition TopS ("  $\top^s$  ") where " $\top^s \equiv p^s \supset^s p^s$ "
14 definition BotS ("  $\bot^s$  ") where " $\bot^s \equiv \neg^s \top^s$ "
15 —<Definition of truth of a formula relative to a model  $\langle W, R, V \rangle$  and a possible world  $w$ >
16 definition RelativeTruthS :: " $\mathcal{W} \Rightarrow \mathcal{R} \Rightarrow \mathcal{V} \Rightarrow w \Rightarrow \sigma \Rightarrow \text{bool}$ " ("  $\models^s$  ") where " $\langle W, R, V \rangle, w \models^s \varphi \equiv \varphi W R V w$ "
17 —<Definition of validity>
18 definition ValS ("  $\models^s$  ") where " $\models^s \varphi \equiv \forall W R V. \forall w:W. \langle W, R, V \rangle, w \models^s \varphi$ "
19 —<Collection of definitions in a bag called DefS>
20 named_theorems DefS declare AtmS_def[DefS,simp] NegS_def[DefS,simp] ImpS_def[DefS,simp]
21 BoxS_def[DefS,simp] OrS_def[DefS,simp] AndS_def[DefS,simp] DiaS_def[DefS,simp] TopS_def[DefS,simp]
22 BotS_def[DefS,simp] RelativeTruthS_def[DefS,simp] ValS_def[DefS,simp]
23 end
```

```
1 theory PMLinHOL_deep imports PMLinHOL_preliminaries (* Christoph Benzmüller, 2025 *)
2 begin
3 —<Deep embedding (of propositional modal logic in HOL)>
4 datatype PML = AtmD  $\mathcal{S}$  ("  $\cdot^d$  ") | NotD PML ("  $\neg^d$  ") | ImpD PML PML (infixr " $\supset^d$ " 93) | BoxD PML ("  $\Box^d$  ")
5 —<Further logical connectives as definitions>
6 definition OrD (infixr " $\vee^d$ " 92) where " $\varphi \vee^d \psi \equiv \neg^d \varphi \supset^d \psi$ "
7 definition AndD (infixr " $\wedge^d$ " 95) where " $\varphi \wedge^d \psi \equiv \neg^d (\varphi \supset^d \neg^d \psi)$ "
8 definition DiaD ("  $\Diamond^d$  ") where " $\Diamond^d \varphi \equiv \neg^d (\Box^d (\neg^d \varphi))$ "
9 definition TopD ("  $\top^d$  ") where " $\top^d \equiv p^d \supset^d p^d$ "
10 definition BotD ("  $\bot^d$  ") where " $\bot^d \equiv \neg^d \top^d$ "
11 —<Definition of truth of a formula relative to a model  $\langle W, R, V \rangle$  and a possible world  $w$ >
12 primrec RelativeTruthD :: " $\mathcal{W} \Rightarrow \mathcal{R} \Rightarrow \mathcal{V} \Rightarrow w \Rightarrow \text{PML} \Rightarrow \text{bool}$ " ("  $\models^d$  ") where
13   " $\langle W, R, V \rangle, w \models^d a^d = (V a w)$ "
14   " $\langle W, R, V \rangle, w \models^d \neg^d \varphi = (\neg \langle W, R, V \rangle, w \models^d \varphi)$ "
15   " $\langle W, R, V \rangle, w \models^d \varphi \supset^d \psi = (\langle W, R, V \rangle, w \models^d \varphi \longrightarrow \langle W, R, V \rangle, w \models^d \psi)$ "
16   " $\langle W, R, V \rangle, w \models^d \Box^d \varphi = (\forall v:W. R w v \longrightarrow \langle W, R, V \rangle, v \models^d \varphi)$ "
17 —<Definition of validity>
18 definition ValD ("  $\models^d$  ") where " $\models^d \varphi \equiv (\forall W R V. \forall w:W. \langle W, R, V \rangle, w \models^d \varphi)$ "
19 —<Collection of definitions in a bag called DefD>
20 named_theorems DefD declare OrD_def[DefD,simp] AndD_def[DefD,simp] DiaD_def[DefD,simp]
21 TopD_def[DefD,simp] BotD_def[DefD,simp] RelativeTruthD_def[DefD,simp] ValD_def[DefD,simp]
22 end
```

deep ($\cdot^d$)

```
1 theory PMLinHOL_shallow_minimal imports PMLinHOL_preliminaries (* Christoph Benzmüller, 2025 *)
2 begin
3 —<The accessibility relation R and the valuation function V are introduced as constants at the meta-level HOL>
4 consts  $R::\mathcal{R}$   $V::\mathcal{V}$ 
5 —<Shallow embedding (of propositional modal logic in HOL)>
6 type_synonym  $\sigma$  = " $w \Rightarrow \text{bool}$ "
7 definition AtmM::" $\sigma \Rightarrow \sigma$ " ("  $\cdot^m$  ") where " $a^m \equiv \lambda w. V a w$ "
8 definition NegM::" $\sigma \Rightarrow \sigma$ " ("  $\neg^m$  ") where " $\neg^m \varphi \equiv \lambda w. \neg \varphi w$ "
9 definition ImpM::" $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixr " $\supset^m$ " 93) where " $\varphi \supset^m \psi \equiv \lambda w. \varphi w \longrightarrow \psi w$ "
10 definition BoxM::" $\sigma \Rightarrow \sigma$ " ("  $\Box^m$  ") where " $\Box^m \varphi \equiv \lambda w. \forall v. R w v \longrightarrow \varphi v$ "
11 —<Further logical connectives as definitions>
12 definition OrM (infixr " $\vee^m$ " 92) where " $\varphi \vee^m \psi \equiv \neg^m \varphi \supset^m \psi$ "
13 definition AndM (infixr " $\wedge^m$ " 95) where " $\varphi \wedge^m \psi \equiv \neg^m (\varphi \supset^m \neg^m \psi)$ "
14 definition DiaM ("  $\Diamond^m$  ") where " $\Diamond^m \varphi \equiv \neg^m (\Box^m (\neg^m \varphi))$ "
15 definition TopM ("  $\top^m$  ") where " $\top^m \equiv p^m \supset^m p^m$ "
16 definition BotM ("  $\bot^m$  ") where " $\bot^m \equiv \neg^m \top^m$ "
17 —<Definition of truth of a formula relative to a model  $\langle W, R, V \rangle$  and a possible world  $w$ >
18 definition RelativeTruthM :: " $w \Rightarrow \sigma \Rightarrow \text{bool}$ " ("  $\models^m$  ") where " $w \models^m \varphi \equiv \varphi w$ "
19 —<Definition of validity>
20 definition ValM ("  $\models^m$  ") where " $\models^m \varphi \equiv \forall w:w. w \models^m \varphi$ "
21 —<Collection of definitions in a bag called DefM>
22 named_theorems DefM declare AtmM_def[DefM,simp] NegM_def[DefM,simp] ImpM_def[DefM,simp]
23 BoxM_def[DefM,simp] OrM_def[DefM,simp] AndM_def[DefM,simp] DiaM_def[DefM,simp] TopM_def[DefM,simp]
24 BotM_def[DefM,simp] RelativeTruthM_def[DefM,simp] ValM_def[DefM,simp]
25 end
```

shallow minimal ($\cdot^m$)

Faithfulness automated

deep ($\cdot^d$)

```
1 theory PMLinHOL_deep imports PMLinHOL_preliminaries (* Christoph Benzmüller, 2025 *)
2 begin
3   <Deep embedding (of propositional modal logic in HOL)>
4   datatype PML = AtmD S ("a^d") | NotD PML ("¬^d") | ImpD PML PML (infix "⊃^d" 93) | BoxD PML ("□^d")
5   <Further logical connectives as definitions>
6   definition OrD (infix "∨^d" 92) where "a ∨^d b ≡ ¬^d (¬^d a ⊃^d ¬^d b)"
7   definition AndD (infix "∧^d" 95) where "a ∧^d b ≡ ¬^d (a ⊃^d ¬^d b)"
8   definition DiaD ("◇^d") where "◇^d a ≡ ¬^d (□^d (¬^d a))"
9   definition TopD ("⊤^d") where "⊤^d ≡ p^d ⊃^d p^d"
10  definition BotD ("⊥^d") where "⊥^d ≡ ¬^d ⊤^d"
11  <Definition of truth of a formula relative to a model <(W,R,V)> and a possible world w>
12  primrec RelativeTruthD :: "W ⇒ R ⇒ V ⇒ w ⇒ PML ⇒ bool" ("⟨_,_,_⟩, w ⊨^d _") where
13    "⟨(W,R,V), w ⊨^d a^d ≡ (V a w)"
14    | "⟨(W,R,V), w ⊨^d ¬^d a^d ≡ (¬ (⟨W,R,V⟩, w ⊨^d a^d))"
15    | "⟨(W,R,V), w ⊨^d a^d ⊃^d b^d ≡ (⟨W,R,V⟩, w ⊨^d a^d ⟶ ⟨W,R,V⟩, w ⊨^d b^d)"
16    | "⟨(W,R,V), w ⊨^d □^d a^d ≡ (∀v. W. R w v ⟶ ⟨W,R,V⟩, v ⊨^d a^d)"
17  <Definition of validity>
18  definition ValD ("⊨^d") where "⊨^d φ ≡ (∀W R V. ∀w. W. ⟨W,R,V⟩, w ⊨^d φ)"
19  <Collection of definitions in a bag called DefD>
20  named_theorems DefD declare OrD_def[DefD,simp] AndD_def[DefD,simp] DiaD_def[DefD,simp]
21  TopD_def[DefD,simp] BotD_def[DefD,simp] RelativeTruthD_def[DefD,simp] ValD_def[DefD,simp]
22 end
```

shallow maximal ($\cdot^s$)

```
1 theory PMLinHOL_shallow imports PMLinHOL_preliminaries (* Christoph Benzmüller, 2025 *)
2 begin
3   <Shallow embedding (of propositional modal logic in HOL)>
4   type_synonym σ = "W ⇒ R ⇒ V ⇒ w ⇒ bool"
5   definition AtmS :: "S ⇒ σ" ("a^s") where "a^s ≡ λW R V w. V a w"
6   definition NegS :: "σ ⇒ σ" ("¬^s") where "¬^s φ ≡ λW R V w. ¬(φ W R V w)"
7   definition ImpS :: "σ ⇒ σ ⇒ σ" (infix "⊃^s" 93) where "φ ⊃^s ψ ≡ λW R V w. (φ W R V w ⟶ (ψ W R V w))"
8   definition BoxS :: "σ ⇒ σ" ("□^s") where "□^s φ ≡ λW R V w. ∀v. W. R w v ⟶ (φ W R V v)"
9   <Further logical connectives as definitions>
10  definition OrS (infix "∨^s" 92) where "a ∨^s b ≡ ¬^s (a ⊃^s ¬^s b)"
11  definition AndS (infix "∧^s" 95) where "a ∧^s b ≡ ¬^s (a ⊃^s ¬^s b)"
12  definition DiaS ("◇^s") where "◇^s a ≡ ¬^s (□^s (¬^s a))"
13  definition TopS ("⊤^s") where "⊤^s ≡ p^s ⊃^s p^s"
14  definition BotS ("⊥^s") where "⊥^s ≡ ¬^s ⊤^s"
15  <Definition of truth of a formula relative to a model <(W,R,V)> and a possible world w>
16  definition RelativeTruthS :: "W ⇒ R ⇒ V ⇒ w ⇒ PML ⇒ bool" ("⟨_,_,_⟩, w ⊨^s _") where "⟨(W,R,V), w ⊨^s φ ≡ φ W R V w"
17  <Definition of validity>
18  definition ValS ("⊨^s") where "⊨^s φ ≡ ∀W R V. ∀w. W. ⟨W,R,V⟩, w ⊨^s φ"
19  <Collection of definitions in a bag called DefS>
20  named_theorems DefS declare AtmS_def[DefS,simp] NegS_def[DefS,simp] ImpS_def[DefS,simp]
21  BoxS_def[DefS,simp] OrS_def[DefS,simp] AndS_def[DefS,simp] DiaS_def[DefS,simp] TopS_def[DefS,simp]
22  BotS_def[DefS,simp] RelativeTruthS_def[DefS,simp] ValS_def[DefS,simp]
23 end
```

shallow minimal ($\cdot^m$)

```
1 theory PMLinHOL_shallow_minimal imports PMLinHOL_preliminaries (* Christoph Benzmüller, 2025 *)
2 begin
3   <The accessibility relation R and the valuation function V are introduced as constants at the meta-level HOL>
4   consts R :: R V :: V
5   <Shallow embedding (of propositional modal logic in HOL)>
6   type_synonym σ = "w ⇒ bool"
7   definition AtmM :: "S ⇒ σ" ("a^m") where "a^m ≡ λw. V a w"
8   definition NegM :: "σ ⇒ σ" ("¬^m") where "¬^m φ ≡ λw. ¬(φ w)"
9   definition ImpM :: "σ ⇒ σ ⇒ σ" (infix "⊃^m" 93) where "φ ⊃^m ψ ≡ λw. φ w ⟶ ψ w"
10  definition BoxM :: "σ ⇒ σ" ("□^m") where "□^m φ ≡ λw. ∀v. R w v ⟶ φ v"
11  <Further logical connectives as definitions>
12  definition OrM (infix "∨^m" 92) where "a ∨^m b ≡ ¬^m (a ⊃^m ¬^m b)"
13  definition AndM (infix "∧^m" 95) where "a ∧^m b ≡ ¬^m (a ⊃^m ¬^m b)"
14  definition DiaM ("◇^m") where "◇^m a ≡ ¬^m (□^m (¬^m a))"
15  definition TopM ("⊤^m") where "⊤^m ≡ p^m ⊃^m p^m"
16  definition BotM ("⊥^m") where "⊥^m ≡ ¬^m ⊤^m"
17  <Definition of truth of a formula relative to a model <(W,R,V)> and a possible world w>
18  definition RelativeTruthM :: "w ⇒ PML ⇒ bool" ("w ⊨^m _") where "w ⊨^m φ ≡ φ w"
19  <Definition of validity>
20  definition ValM ("⊨^m") where "⊨^m φ ≡ ∀w. w ⊨^m φ"
21  <Collection of definitions in a bag called DefM>
22  named_theorems DefM declare AtmM_def[DefM,simp] NegM_def[DefM,simp] ImpM_def[DefM,simp]
23  BoxM_def[DefM,simp] OrM_def[DefM,simp] AndM_def[DefM,simp] DiaM_def[DefM,simp] TopM_def[DefM,simp]
24  BotM_def[DefM,simp] RelativeTruthM_def[DefM,simp] ValM_def[DefM,simp]
25 end
```

```
1 theory PMLinHOL_faithfulness imports PMLinHOL_deep PMLinHOL_shallow PMLinHOL_shallow_minimal (* C.B., 2025 *)
2 begin
3   <Mappings: deep to maximal shallow and deep to minimal shallow>
4   primrec DpToShMax ("[]") where "[a^d] = a^s" | "[¬^d φ] = ¬^s [φ]" | "[φ ⊃^d ψ] = [φ] ⊃^s [ψ]" | "[□^d φ] = □^s [φ]"
5   primrec DpToShMin ("[]") where "[a^d] = a^m" | "[¬^d φ] = ¬^m [φ]" | "[φ ⊃^d ψ] = [φ] ⊃^m [ψ]" | "[□^d φ] = □^m [φ]"
6   <Proving faithfulness between deep and maximal shallow>
7   theorem Faithful1a: "∀W R V. ∀w. W. ⟨W,R,V⟩, w ⊨^d φ ⟷ ⟨W,R,V⟩, w ⊨^s [φ]"
8   theorem Faithful1b: "⊨^d φ ⟷ ⊨^s [φ]"
9   <Proving faithfulness between deep and minimal shallow>
10  theorem Faithful2: "∀w. ⟨(λx::w. True),R,V⟩, w ⊨^d φ ⟷ w ⊨^m [φ]"
11  <Proving faithfulness maximal shallow and minimal shallow>
12  theorem Faithful3: "∀w. ⟨(λx::w. True),R,V⟩, w ⊨^s [φ] ⟷ w ⊨^m [φ]"
13  <Additional check for soundness for the minimal shallow embedding>
14  lemma Sound1: "⊨^m ψ ⟷ (∃φ. ψ = [φ] ∧ ⊨^d φ)" by (smt Faithful2 DefM DefD RelativeTruthD.simps ext[of ψ "[x ⊃^d x]"])
15  lemma Sound2: "⊨^m ψ ⟷ (∃φ. ψ = [φ] ∧ ⊨^m [φ])" using Sound1 by blast
16 end
```

apply induct by auto
using Faithful1a by auto

apply induct by auto

apply induct by auto

Faithfulness automated

$\langle \cdot \rangle$ - Mapping of deep formulas to maximal shallow formulas

$\llbracket \cdot \rrbracket$ - Mapping of deep formulas to minimal shallow formulas

```
1 theory PMLinHOL_faithfulness imports PMLinHOL_deep PMLinHOL_shallow PMLinHOL_shallow_minimal (* C.B., 2025 *)
2 begin
3 —<Mappings: deep to maximal shallow and deep to minimal shallow>
4 primrec DpToShMax (" $\langle \cdot \rangle$ ") where " $\langle a^d \rangle = a^s$ " | " $\langle \neg^d \varphi \rangle = \neg^s \langle \varphi \rangle$ " | " $\langle \varphi \supset^d \psi \rangle = \langle \varphi \rangle \supset^s \langle \psi \rangle$ " | " $\langle \Box^d \varphi \rangle = \Box^s \langle \varphi \rangle$ "
5 primrec DpToShMin (" $\llbracket \cdot \rrbracket$ ") where " $\llbracket a^d \rrbracket = a^m$ " | " $\llbracket \neg^d \varphi \rrbracket = \neg^m \llbracket \varphi \rrbracket$ " | " $\llbracket \varphi \supset^d \psi \rrbracket = \llbracket \varphi \rrbracket \supset^m \llbracket \psi \rrbracket$ " | " $\llbracket \Box^d \varphi \rrbracket = \Box^m \llbracket \varphi \rrbracket$ "
6 —<Proving faithfulness between deep and maximal shallow>
7 theorem Faithful1a: " $\forall W R V. \forall w:W. \langle W, R, V \rangle, w \models^d \varphi \longleftrightarrow \langle W, R, V \rangle, w \models^s \langle \varphi \rangle$ " apply induct by auto
8 theorem Faithful1b: " $\models^d \varphi \longleftrightarrow \models^s \langle \varphi \rangle$ " using Faithful1a by auto
9 —<Proving faithfulness between deep and minimal shallow>
10 theorem Faithful2: " $\forall w. \langle (\lambda x::w. \text{True}), R, V \rangle, w \models^d \varphi \longleftrightarrow w \models^m \llbracket \varphi \rrbracket$ " apply induct by auto
11 —<Proving faithfulness maximal shallow and minimal shallow>
12 theorem Faithful3: " $\forall w. \langle (\lambda x::w. \text{True}), R, V \rangle, w \models^s \langle \varphi \rangle \longleftrightarrow w \models^m \llbracket \varphi \rrbracket$ " apply induct by auto
13 —<Additional check for soundness for the minimal shallow embedding>
14 lemma Sound1: " $\models^m \psi \longleftrightarrow (\exists \varphi. \psi = \llbracket \varphi \rrbracket \wedge \models^d \varphi)$ " by (smt Faithful2 DefM DefD RelativeTruthD.simps ext[of  $\psi$  " $\llbracket x \supset^d x \rrbracket$ "]])
15 lemma Sound2: " $\models^m \psi \longleftrightarrow (\exists \varphi. \psi = \llbracket \varphi \rrbracket \wedge \models^m \llbracket \varphi \rrbracket)$ " using Sound1 by blast
16 end
```


Faithfulness automated

A deep formula is valid if and only if its maximal shallow translation is valid.

```
1 theory PMLinHOL_faithfulness imports PMLinHOL_deep PMLinHOL_shallow PMLinHOL_shallow_minimal (* C.B., 2025 *)
2 begin
3 —<Mappings: deep to maximal shallow and deep to minimal shallow>
4 primrec DpToShMax ("⟦_⟧") where " $\langle a^d \rangle = a^s$ " | " $\langle \neg^d \varphi \rangle = \neg^s \langle \varphi \rangle$ " | " $\langle \varphi \supset^d \psi \rangle = \langle \varphi \rangle \supset^s \langle \psi \rangle$ " | " $\langle \Box^d \varphi \rangle = \Box^s \langle \varphi \rangle$ "
5 primrec DpToShMin ("⟦_⟧") where " $\llbracket a^d \rrbracket = a^m$ " | " $\llbracket \neg^d \varphi \rrbracket = \neg^m \llbracket \varphi \rrbracket$ " | " $\llbracket \varphi \supset^d \psi \rrbracket = \llbracket \varphi \rrbracket \supset^m \llbracket \psi \rrbracket$ " | " $\llbracket \Box^d \varphi \rrbracket = \Box^m \llbracket \varphi \rrbracket$ "
6 —<Proving faithfulness between deep and maximal shallow>
7 theorem Faithful1a: " $\forall W R V. \forall w:W. \langle W, R, V \rangle, w \models^d \varphi \longleftrightarrow \langle W, R, V \rangle, w \models^s \langle \varphi \rangle$ " apply induct by auto
8 theorem Faithful1b: " $\models^d \varphi \longleftrightarrow \models^s \langle \varphi \rangle$ " using Faithful1a by auto
9 —<Proving faithfulness between deep and minimal shallow>
10 theorem Faithful2: " $\forall w. \langle (\lambda x::w. \text{True}), R, V \rangle, w \models^d \varphi \longleftrightarrow w \models^m \llbracket \varphi \rrbracket$ " apply induct by auto
11 —<Proving faithfulness maximal shallow and minimal shallow>
12 theorem Faithful3: " $\forall w. \langle (\lambda x::w. \text{True}), R, V \rangle, w \models^s \langle \varphi \rangle \longleftrightarrow w \models^m \llbracket \varphi \rrbracket$ " apply induct by auto
13 —<Additional check for soundness for the minimal shallow embedding>
14 lemma Sound1: " $\models^m \psi \longleftrightarrow (\exists \varphi. \psi = \llbracket \varphi \rrbracket \wedge \models^d \varphi)$ " by (smt Faithful2 DefM DefD RelativeTruthD.simps ext[of  $\psi$  " $\llbracket x \supset^d x \rrbracket$ "]])
15 lemma Sound2: " $\models^m \psi \longleftrightarrow (\exists \varphi. \psi = \llbracket \varphi \rrbracket \wedge \models^m \llbracket \varphi \rrbracket)$ " using Sound1 by blast
16 end
```


Faithfulness automated

Similar relative faithfulness theorems for the

- deep and minimal shallow embedding, and for the
- maximal shallow and minimal shallow embedding

```
1 theory PMLinHOL_faithfulness imports PMLinHOL_deep PMLinHOL_shallow PMLinHOL_shallow_minimal (* C.B., 2025 *)
2 begin
3 —<Mappings: deep to maximal shallow and deep to minimal shallow>
4 primrec DpToShMax ("⟦_⟧") where " $\langle a^d \rangle = a^s$ " | " $\langle \neg^d \varphi \rangle = \neg^s \langle \varphi \rangle$ " | " $\langle \varphi \supset^d \psi \rangle = \langle \varphi \rangle \supset^s \langle \psi \rangle$ " | " $\langle \Box^d \varphi \rangle = \Box^s \langle \varphi \rangle$ "
5 primrec DpToShMin ("⟦_⟧") where " $\langle a^d \rangle = a^m$ " | " $\langle \neg^d \varphi \rangle = \neg^m \langle \varphi \rangle$ " | " $\langle \varphi \supset^d \psi \rangle = \langle \varphi \rangle \supset^m \langle \psi \rangle$ " | " $\langle \Box^d \varphi \rangle = \Box^m \langle \varphi \rangle$ "
6 —<Proving faithfulness between deep and maximal shallow>
7 theorem Faithful1a: " $\forall W R V. \forall w:W. \langle W, R, V \rangle, w \models^d \varphi \longleftrightarrow \langle W, R, V \rangle, w \models^s \langle \varphi \rangle$ " apply induct by auto
8 theorem Faithful1b: " $\models^d \varphi \longleftrightarrow \models^s \langle \varphi \rangle$ " using Faithful1a by auto
9 —<Proving faithfulness between deep and minimal shallow>
10 theorem Faithful2: " $\forall w. \langle (\lambda x::w. \text{True}), R, V \rangle, w \models^d \varphi \longleftrightarrow w \models^m \langle \varphi \rangle$ " apply induct by auto
11 —<Proving faithfulness maximal shallow and minimal shallow>
12 theorem Faithful3: " $\forall w. \langle (\lambda x::w. \text{True}), R, V \rangle, w \models^s \langle \varphi \rangle \longleftrightarrow w \models^m \langle \varphi \rangle$ " apply induct by auto
13 —<Additional check for soundness for the minimal shallow embedding>
14 lemma Sound1: " $\models^m \psi \longleftrightarrow (\exists \varphi. \psi = \langle \varphi \rangle \wedge \models^d \varphi)$ " by (smt Faithful2 DefM DefD RelativeTruthD.simps ext[of  $\psi$  " $\langle x \supset^d x \rangle$ "]])
15 lemma Sound2: " $\models^m \psi \longleftrightarrow (\exists \varphi. \psi = \langle \varphi \rangle \wedge \models^m \langle \varphi \rangle)$ " using Sound1 by blast
16 end
```


Experiments

Conducted with all embeddings — but presented here for the minimal shallow embedding

Experiments

```
1 theory PMLinHOL_shallow_minimal_tests imports PMLinHOL_shallow_minimal (* Christoph Benz Müller, 2025 *)  
2 begin  
3 —< Hilbert calculus: proving that the schematic axioms and rules are implied by the embedding >  
4 lemma H1: " $\models^m \varphi \supset^m (\psi \supset^m \varphi)$ " by auto  
5 lemma H2: " $\models^m (\varphi \supset^m (\psi \supset^m \gamma)) \supset^m ((\varphi \supset^m \psi) \supset^m (\varphi \supset^m \gamma))$ " by auto  
6 lemma H3: " $\models^m (\neg^m \varphi \supset^m \neg^m \psi) \supset^m (\psi \supset^m \varphi)$ " by auto  
7 lemma MP: " $\models^m \varphi \implies \models^m (\varphi \supset^m \psi) \implies \models^m \psi$ " by auto
```

Hilbert axioms and modus ponens rule are implied

H1: $\models \varphi \supset (\psi \supset \varphi)$

H2: $\models (\varphi \supset (\psi \supset \gamma)) \supset ((\varphi \supset \psi) \supset (\varphi \supset \gamma))$

H3: $\models (\neg \varphi \supset \neg \psi) \supset (\psi \supset \varphi)$

MP: $\models \varphi \implies \models \varphi \supset \psi \implies \models \psi$

Experiments

```
1 theory PMLinHOL_shallow_minimal_tests imports PMLinHOL_shallow_minimal (* Christoph Benz Müller, 2025 *)  
2 begin  
3 —< Hilbert calculus: proving that the schematic axioms and rules are implied by the embedding >  
4 lemma H1: " $\models^m \varphi \supset^m (\psi \supset^m \varphi)$ " by auto  
5 lemma H2: " $\models^m (\varphi \supset^m (\psi \supset^m \gamma)) \supset^m ((\varphi \supset^m \psi) \supset^m (\varphi \supset^m \gamma))$ " by auto  
6 lemma H3: " $\models^m (\neg^m \varphi \supset^m \neg^m \psi) \supset^m (\psi \supset^m \varphi)$ " by auto  
7 lemma MP: " $\models^m \varphi \implies \models^m (\varphi \supset^m \psi) \implies \models^m \psi$ " by auto  
8 —< Reasoning with the Hilbert calculus: interactive and fully automated >  
9 lemma HCderived1: " $\models^m (\varphi \supset^m \varphi)$ " —< sledgehammer(H1 H2 H3 MP) returns: by (metis H1 H2 MP) >
```

Simple automated proof of $\varphi \supset \varphi$ at level of this Hilbert calculus

Experiments

```
1 theory PMLinHOL_shallow_minimal_tests imports PMLinHOL_shallow_minimal (* Christoph Benz Müller, 2025 *)
2 begin
3 —< Hilbert calculus: proving that the schematic axioms and rules are implied by the embedding >
4 lemma H1: " $\models^m \varphi \supset^m (\psi \supset^m \varphi)$ " by auto
5 lemma H2: " $\models^m (\varphi \supset^m (\psi \supset^m \gamma)) \supset^m ((\varphi \supset^m \psi) \supset^m (\varphi \supset^m \gamma))$ " by auto
6 lemma H3: " $\models^m (\neg^m \varphi \supset^m \neg^m \psi) \supset^m (\psi \supset^m \varphi)$ " by auto
7 lemma MP: " $\models^m \varphi \implies \models^m (\varphi \supset^m \psi) \implies \models^m \psi$ " by auto
8 —< Reasoning with the Hilbert calculus: interactive and fully automated >
9 lemma HCderived1: " $\models^m (\varphi \supset^m \varphi)$ " —< sledgehammer(H1 H2 H3 MP) returns: by (metis H1 H2 MP) >
10 proof - have 1: " $\models^m \varphi \supset^m ((\psi \supset^m \varphi) \supset^m \varphi)$ " using H1 by auto
11           have 2: " $\models^m (\varphi \supset^m ((\psi \supset^m \varphi) \supset^m \varphi)) \supset^m ((\varphi \supset^m (\psi \supset^m \varphi)) \supset^m (\varphi \supset^m \varphi))$ " using H2 by auto
12           have 3: " $\models^m (\varphi \supset^m (\psi \supset^m \varphi)) \supset^m (\varphi \supset^m \varphi)$ " using 1 2 MP by meson
13           have 4: " $\models^m \varphi \supset^m (\psi \supset^m \varphi)$ " using H1 by auto
14           thus ?thesis using 3 4 MP by meson qed
```

Interactive proof of $\varphi \supset \varphi$ in this Hilbert calculus

Experiments

```
1 theory PMLinHOL_shallow_minimal_tests imports PMLinHOL_shallow_minimal (* Christoph Benz Müller, 2025 *)  
2 begin  
3 —< Hilbert calculus: proving that the schematic axioms and rules are implied by the embedding >  
4 lemma H1: " $\models^m \varphi \supset^m (\psi \supset^m \varphi)$ " by auto  
5 lemma H2: " $\models^m (\varphi \supset^m (\psi \supset^m \gamma)) \supset^m ((\varphi \supset^m \psi) \supset^m (\varphi \supset^m \gamma))$ " by auto  
6 lemma H3: " $\models^m (\neg^m \varphi \supset^m \neg^m \psi) \supset^m (\psi \supset^m \varphi)$ " by auto  
7 lemma MP: " $\models^m \varphi \implies \models^m (\varphi \supset^m \psi) \implies \models^m \psi$ " by auto  
8 —< Reasoning with the Hilbert calculus: interactive and fully automated >  
9 lemma HCderived1: " $\models^m (\varphi \supset^m \varphi)$ " —< sledgehammer(H1 H2 H3 MP) returns: by (metis H1 H2 MP) >  
10 proof - have 1: " $\models^m \varphi \supset^m ((\psi \supset^m \varphi) \supset^m \varphi)$ " using H1 by auto  
11 have 2: " $\models^m (\varphi \supset^m ((\psi \supset^m \varphi) \supset^m \varphi)) \supset^m ((\varphi \supset^m (\psi \supset^m \varphi)) \supset^m (\varphi \supset^m \varphi))$ " using H2 by auto  
12 have 3: " $\models^m (\varphi \supset^m (\psi \supset^m \varphi)) \supset^m (\varphi \supset^m \varphi)$ " using 1 2 MP by meson  
13 have 4: " $\models^m \varphi \supset^m (\psi \supset^m \varphi)$ " using H1 by auto  
14 thus ?thesis using 3 4 MP by meson qed  
15 lemma HCderived2: " $\models^m \varphi \supset^m (\neg^m \varphi \supset^m \psi)$ " by (metis H1 H2 H3 MP)  
16 lemma HCderived3: " $\models^m (\neg^m \varphi \supset^m \varphi) \supset^m \varphi$ " by (metis H1 H2 H3 MP)  
17 lemma HCderived4: " $\models^m (\varphi \supset^m \psi) \supset^m (\neg^m \psi \supset^m \neg^m \varphi)$ " by auto
```

Further automated proofs of simple example problems

Experiments

```
1 theory PMLinHOL_shallow_minimal_tests imports PMLinHOL_shallow_minimal (* Christoph Benzmüller, 2025 *)
2 begin
3 —< Hilbert calculus: proving that the schematic axioms and rules are implied by the embedding >
4 lemma H1: " $\models^m \varphi \supset^m (\psi \supset^m \varphi)$ " by auto
5 lemma H2: " $\models^m (\varphi \supset^m (\psi \supset^m \gamma)) \supset^m ((\varphi \supset^m \psi) \supset^m (\varphi \supset^m \gamma))$ " by auto
6 lemma H3: " $\models^m (\neg^m \varphi \supset^m \neg^m \psi) \supset^m (\psi \supset^m \varphi)$ " by auto
7 lemma MP: " $\models^m \varphi \implies \models^m (\varphi \supset^m \psi) \implies \models^m \psi$ " by auto
8 —< Reasoning with the Hilbert calculus: interactive and fully automated >
9 lemma HCderived1: " $\models^m (\varphi \supset^m \varphi)$ " —< sledgehammer(H1 H2 H3 MP) returns: by (metis H1 H2 MP) >
10 proof - have 1: " $\models^m \varphi \supset^m ((\psi \supset^m \varphi) \supset^m \varphi)$ " using H1 by auto
11           have 2: " $\models^m (\varphi \supset^m ((\psi \supset^m \varphi) \supset^m \varphi)) \supset^m ((\varphi \supset^m (\psi \supset^m \varphi)) \supset^m (\varphi \supset^m \varphi))$ " using H2 by auto
12           have 3: " $\models^m (\varphi \supset^m (\psi \supset^m \varphi)) \supset^m (\varphi \supset^m \varphi)$ " using 1 2 MP by meson
13           have 4: " $\models^m \varphi \supset^m (\psi \supset^m \varphi)$ " using H1 by auto
14           thus ?thesis using 3 4 MP by meson qed
15 lemma HCderived2: " $\models^m \varphi \supset^m (\neg^m \varphi \supset^m \psi)$ " by (metis H1 H2 H3 MP)
16 lemma HCderived3: " $\models^m (\neg^m \varphi \supset^m \varphi) \supset^m \varphi$ " by (metis H1 H2 H3 MP)
17 lemma HCderived4: " $\models^m (\varphi \supset^m \psi) \supset^m (\neg^m \psi \supset^m \neg^m \varphi)$ " by auto
18 —< Modal logic: the schematic necessitation rule and distribution axiom are implied >
19 lemma Nec: " $\models^m \varphi \implies \models^m \Box^m \varphi$ " by (smt DefM)
20 lemma Dist: " $\models^m \Box^m (\varphi \supset^m \psi) \supset^m (\Box^m \varphi \supset^m \Box^m \psi)$ " by auto
```

Necessitation rule and distribution axiom are implied

Nec: $\models \varphi \implies \models \Box \varphi$

Dist: $\models \Box (\varphi \supset \psi) \supset (\Box \varphi \supset \Box \psi)$

Experiments

```
1 theory PMLinHOL_shallow_minimal_tests imports PMLinHOL_shallow_minimal (* Christoph Benzmüller, 2025 *)
2 begin
3 —< Hilbert calculus: proving that the schematic axioms and rules are implied by the embedding >
4 lemma H1: " $\models^m \varphi \supset^m (\psi \supset^m \varphi)$ " by auto
5 lemma H2: " $\models^m (\varphi \supset^m (\psi \supset^m \gamma)) \supset^m ((\varphi \supset^m \psi) \supset^m (\varphi \supset^m \gamma))$ " by auto
6 lemma H3: " $\models^m (\neg^m \varphi \supset^m \neg^m \psi) \supset^m (\psi \supset^m \varphi)$ " by auto
7 lemma MP: " $\models^m \varphi \implies \models^m (\varphi \supset^m \psi) \implies \models^m \psi$ " by auto
8 —< Reasoning with the Hilbert calculus: interactive and fully automated >
9 lemma HCderived1: " $\models^m (\varphi \supset^m \varphi)$ " —< sledgehammer(H1 H2 H3 MP) returns: by (metis H1 H2 MP) >
10 proof - have 1: " $\models^m \varphi \supset^m ((\psi \supset^m \varphi) \supset^m \varphi)$ " using H1 by auto
11 have 2: " $\models^m (\varphi \supset^m ((\psi \supset^m \varphi) \supset^m \varphi)) \supset^m ((\varphi \supset^m (\psi \supset^m \varphi)) \supset^m (\varphi \supset^m \varphi))$ " using H2 by auto
12 have 3: " $\models^m (\varphi \supset^m (\psi \supset^m \varphi)) \supset^m (\varphi \supset^m \varphi)$ " using 1 2 MP by meson
13 have 4: " $\models^m \varphi \supset^m (\psi \supset^m \varphi)$ " using H1 by auto
14 thus ?thesis using 3 4 MP by meson qed
15 lemma HCderived2: " $\models^m \varphi \supset^m (\neg^m \varphi \supset^m \psi)$ " by (metis H1 H2 H3 MP)
16 lemma HCderived3: " $\models^m (\neg^m \varphi \supset^m \varphi) \supset^m \varphi$ " by (metis H1 H2 H3 MP)
17 lemma HCderived4: " $\models^m (\varphi \supset^m \psi) \supset^m (\neg^m \psi \supset^m \neg^m \varphi)$ " by auto
18 —< Modal logic: the schematic necessitation rule and distribution axiom are implied >
19 lemma Nec: " $\models^m \varphi \implies \models^m \Box^m \varphi$ " by (smt DefM)
20 lemma Dist: " $\models^m \Box^m (\varphi \supset^m \psi) \supset^m (\Box^m \varphi \supset^m \Box^m \psi)$ " by auto
21 —< Correspondence theory: correct statements >
22 lemma cM: "reflexive R  $\longleftrightarrow (\forall \varphi. \models^m \Box^m \varphi \supset^m \varphi)$ " by auto
23 lemma cBa: "symmetric R  $\longrightarrow (\forall \varphi. \models^m \varphi \supset^m \Box^m \Diamond^m \varphi)$ " by auto
24 lemma cBb: "symmetric R  $\longleftarrow (\forall \varphi. \models^m \varphi \supset^m \Box^m \Diamond^m \varphi)$ " by (metis DefM)
25 lemma c4a: "transitive R  $\longrightarrow (\forall \varphi. \models^m \Box^m \varphi \supset^m \Box^m (\Box^m \varphi))$ " by (smt DefM)
26 lemma c4b: "transitive R  $\longleftarrow (\forall \varphi. \models^m \Box^m \varphi \supset^m \Box^m (\Box^m \varphi))$ " by auto
```

Sahlqvist correspondences automated

Experiments

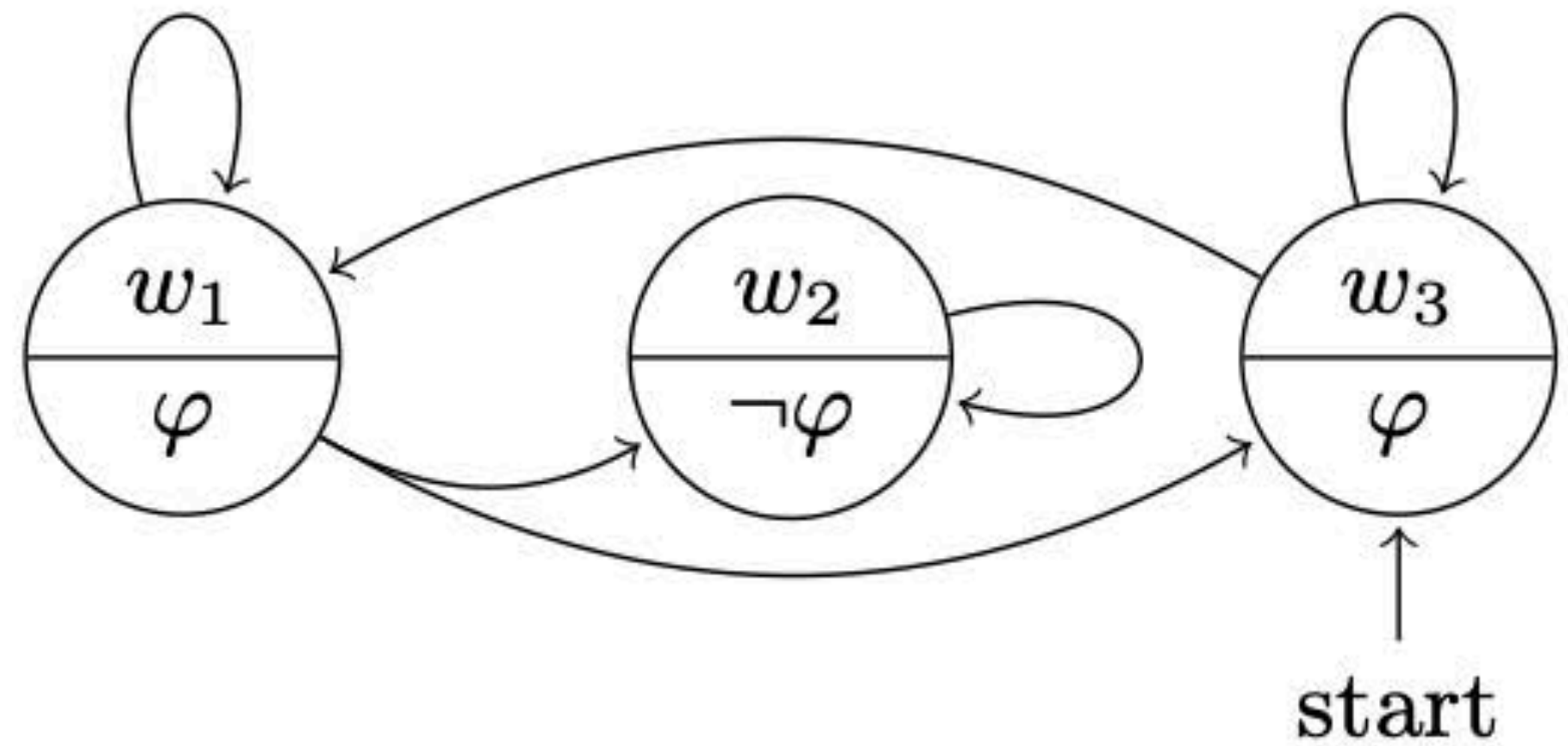
```
1 theory PMLinHOL_shallow_minimal_tests imports PMLinHOL_shallow_minimal (* Christoph Benzmüller, 2025 *)
2 begin
3 —< Hilbert calculus: proving that the schematic axioms and rules are implied by the embedding >
4 lemma H1: " $\models^m \varphi \supset^m (\psi \supset^m \varphi)$ " by auto
5 lemma H2: " $\models^m (\varphi \supset^m (\psi \supset^m \gamma)) \supset^m ((\varphi \supset^m \psi) \supset^m (\varphi \supset^m \gamma))$ " by auto
6 lemma H3: " $\models^m (\neg^m \varphi \supset^m \neg^m \psi) \supset^m (\psi \supset^m \varphi)$ " by auto
7 lemma MP: " $\models^m \varphi \implies \models^m (\varphi \supset^m \psi) \implies \models^m \psi$ " by auto
8 —< Reasoning with the Hilbert calculus: interactive and fully automated >
9 lemma HCderived1: " $\models^m (\varphi \supset^m \varphi)$ " —< sledgehammer(H1 H2 H3 MP) returns: by (metis H1 H2 MP) >
10 proof - have 1: " $\models^m \varphi \supset^m ((\psi \supset^m \varphi) \supset^m \varphi)$ " using H1 by auto
11           have 2: " $\models^m (\varphi \supset^m ((\psi \supset^m \varphi) \supset^m \varphi)) \supset^m ((\varphi \supset^m (\psi \supset^m \varphi)) \supset^m (\varphi \supset^m \varphi))$ " using H2 by auto
12           have 3: " $\models^m (\varphi \supset^m (\psi \supset^m \varphi)) \supset^m (\varphi \supset^m \varphi)$ " using 1 2 MP by meson
13           have 4: " $\models^m \varphi \supset^m (\psi \supset^m \varphi)$ " using H1 by auto
14           thus ?thesis using 3 4 MP by meson qed
15 lemma HCderived2: " $\models^m \varphi \supset^m (\neg^m \varphi \supset^m \psi)$ " by (metis H1 H2 H3 MP)
16 lemma HCderived3: " $\models^m (\neg^m \varphi \supset^m \varphi) \supset^m \varphi$ " by (metis H1 H2 H3 MP)
17 lemma HCderived4: " $\models^m (\varphi \supset^m \psi) \supset^m (\neg^m \psi \supset^m \neg^m \varphi)$ " by auto
18 —< Modal logic: the schematic necessitation rule and distribution axiom are implied >
19 lemma Nec: " $\models^m \varphi \implies \models^m \Box^m \varphi$ " by (smt DefM)
20 lemma Dist: " $\models^m \Box^m (\varphi \supset^m \psi) \supset^m (\Box^m \varphi \supset^m \Box^m \psi)$ " by auto
21 —< Correspondence theory: correct statements >
22 lemma cM: "reflexive R  $\longleftrightarrow (\forall \varphi. \models^m \Box^m \varphi \supset^m \varphi)$ " by auto
23 lemma cBa: "symmetric R  $\longrightarrow (\forall \varphi. \models^m \varphi \supset^m \Box^m \Diamond^m \varphi)$ " by auto
24 lemma cBb: "symmetric R  $\longleftarrow (\forall \varphi. \models^m \varphi \supset^m \Box^m \Diamond^m \varphi)$ " by (metis DefM)
25 lemma c4a: "transitive R  $\longrightarrow (\forall \varphi. \models^m \Box^m \varphi \supset^m \Box^m (\Box^m \varphi))$ " by (smt DefM)
26 lemma c4b: "transitive R  $\longleftarrow (\forall \varphi. \models^m \Box^m \varphi \supset^m \Box^m (\Box^m \varphi))$ " by auto
27 —< Correspondence theory: incorrect statements >
28 lemma "reflexive R  $\longrightarrow (\forall \varphi. \models^m \Box^m \varphi \supset^m \Box^m (\Box^m \varphi))$ " nitpick[card w=3, show_all] oops —< nitpick: Counterexample >
```

Incorrect correspondence claim

$$\text{reflexive } R \longrightarrow (\forall \varphi. \models \Box \varphi \supset \Box \Box \varphi)$$

Experiments

```
1 theory PMLinHOL_shallow_minimal_tests imports PMLinHOL_shallow_minimal (* Christoph Benzmüller, 2025 *)
2 begin
3 —< Hilbert calculus: proving that the schematic axioms and rules are implied by the embedding >
4 lemma H1: "⊨m φ ⊃m (ψ ⊃m φ)" by auto
5 lemma H2: "⊨m (φ ⊃m (ψ ⊃m γ)) ⊃m ((φ ⊃m ψ) ⊃m (φ ⊃m γ))" by auto
6 lemma H3: "⊨m (¬m φ ⊃m ¬m ψ) ⊃m (ψ ⊃m φ)" by auto
7 lemma MP: "⊨m φ ⇒ ⊨m (φ ⊃m ψ) ⇒ ⊨m ψ" by auto
```



Nitpick found a counterexample for card w = 3:
Skolem constants:
??BoxM.v = w₁
??BoxM.v = w₂
??ValM.w = w₃
φ = (λx. _)(w₁ := True, w₂ := False, w₃ := True)
Constant:
R =
(λx. _)
(w₁ := (λx. _)(w₁ := True, w₂ := True, w₃ := True),
w₂ := (λx. _)(w₁ := False, w₂ := True, w₃ := False),
w₃ := (λx. _)(w₁ := True, w₂ := False, w₃ := True))

Fig. 5. Counterexample reported by nitpick in line 28 of Fig. 4

[See paper at LPLR 2025]

```
27 —< Correspondence theory: incorrect statements >
28 lemma "reflexive R → (∀φ. ⊨m □m φ ⊃m □m (□m φ))" nitpick[card w=3, show_all] oops —< nitpick: Counterexample >
```

Incorrect correspondence claim — counterexample by Nitpick

$$\text{reflexive } R \longrightarrow (\forall \varphi . \models \Box \varphi \supset \Box \Box \varphi)$$

Experiments

modal reasoning

```
1 theory PMLinHOL_shallow_minimal_further_tests imports PMLinHOL_shallow_minimal_tests (* C.B., 2025 *)
2 begin
3 —<Implied modal principle>
4 lemma K_Dia: " $\models^m (\Box^m (\varphi \supset^m \psi)) \supset^m ((\Diamond^m \varphi) \supset^m \Diamond^m \psi)$ " by auto
5 —<Example 6.10 of Sider (2009) Logic for Philosophy>
6 lemma T1a: " $\models^m \Box^m p^m \supset^m ((\Diamond^m q^m) \supset^m \Diamond^m (p^m \wedge^m q^m))$ " by auto —<fast automation in meta-logic HOL>
7 lemma T1b: " $\models^m \Box^m p^m \supset^m ((\Diamond^m q^m) \supset^m \Diamond^m (p^m \wedge^m q^m))$ "
8   proof - —<alternative interactive proof in modal object logic K>
9     have 1: " $\models^m p^m \supset^m (q^m \supset^m (p^m \wedge^m q^m))$ " unfolding AndM_def using H1 H2 H3 MP by metis
10    have 2: " $\models^m \Box^m (p^m \supset^m (q^m \supset^m (p^m \wedge^m q^m)))$ " using 1 Nec by metis
11    have 3: " $\models^m \Box^m p^m \supset^m \Box^m (q^m \supset^m (p^m \wedge^m q^m))$ " using 2 Dist MP by metis
12    have 4: " $\models^m (\Box^m (q^m \supset^m (p^m \wedge^m q^m))) \supset^m ((\Diamond^m q^m) \supset^m \Diamond^m (p^m \wedge^m q^m))$ " using K_Dia by metis
13    have 5: " $\models^m \Box^m p^m \supset^m ((\Diamond^m q^m) \supset^m \Diamond^m (p^m \wedge^m q^m))$ " using 3 4 H1 H2 MP by metis
14    thus ?thesis . qed
15 end
```

inspecting, and reasoning with, the Löb formula

```
1 theory PMLinHOL_shallow_minimal_Loeb_tests imports PMLinHOL_shallow_minimal
2 begin
3 —<Löb axiom: with the minimal shallow embedding automated reasoning tools are still partly responsive>
4 lemma Loeb1M: " $\forall \varphi. \models^m \Box^m (\Box^m \varphi \supset^m \varphi) \supset^m \Box^m \varphi$ " nitpick [card w=1, card S=1] oops —<nitpick: Counterexample.>
5 lemma Loeb2M: "(conversewf R  $\wedge$  transitive R)  $\longrightarrow$  ( $\forall \varphi. \models^m \Box^m (\Box^m \varphi \supset^m \varphi) \supset^m \Box^m \varphi$ )"
6   by (smt (verit, del_insts) BoxM_def ImpM_def RelativeTruthM_def ValM_def)
7 lemma Loeb3aM: "conversewf R  $\longleftarrow$  ( $\forall \varphi. \models^m \Box^m (\Box^m \varphi \supset^m \varphi) \supset^m \Box^m \varphi$ )" unfolding DefM by blast
8 lemma Loeb3bM: "transitive R  $\longleftarrow$  ( $\forall \varphi. \models^m \Box^m (\Box^m \varphi \supset^m \varphi) \supset^m \Box^m \varphi$ )" —<sledgehammer: no proof>
9   proof assume 1: " $\forall \varphi. \models^m \Box^m (\Box^m \varphi \supset^m \varphi) \supset^m \Box^m \varphi$ "
10    have 2: " $\forall \varphi. \models^m \Box^m \varphi \supset^m \Box^m ((\Box^m (\Box^m \varphi \wedge^m \varphi)) \supset^m ((\Box^m \varphi) \wedge^m \varphi))$ " by auto
11    show "transitive R" by (smt (z3) 1 2 DefM (10,11,2,3,4,6)) qed
12 lemma Loeb3M: "(conversewf R  $\wedge$  transitive R)  $\longleftarrow$  ( $\forall \varphi. \models^m \Box^m (\Box^m \varphi \supset^m \varphi) \supset^m \Box^m \varphi$ )"
13   using Loeb3aM Loeb3bM by force
14 end
```


PML Evaluation

(new — extending results from CADE 2025)

Evaluation

$F_1: \Diamond\Diamond\varphi \supset \Diamond\varphi$	$F_6: ((\Box(\varphi \supset \psi)) \wedge \Diamond\Box\neg\psi) \supset \neg\Diamond\psi$
$F_2: \Diamond\Box\varphi \supset \Box\Diamond\varphi$	$F_7: \Diamond\varphi \supset \Box(\varphi \vee \Diamond\varphi)$
$F_3: \Diamond\Box\varphi \supset \Box\varphi$	$F_8: \Diamond(\Box\varphi) \supset (\varphi \vee \Diamond\varphi)$
$F_4: \Box\Diamond\Box\Diamond\varphi \supset \Box\Diamond\varphi$	$F_9: (\Box\Diamond\varphi \wedge \Box\Diamond\neg\varphi) \supset \Diamond\Diamond\varphi$
$F_5: \Diamond(\varphi \wedge \Diamond\psi) \supset (\Diamond\varphi \wedge \Diamond\psi)$	$F_{10}: (\Box(\varphi \supset \Box\psi) \wedge \Box\Diamond\neg\psi) \supset \neg\Box\psi$

$AxT:$	$\forall\varphi. \models \Box\varphi \supset \varphi$
$AxB:$	$\forall\varphi. \models \varphi \supset \Box\Diamond\varphi$
$Ax4:$	$\forall\varphi. \models \Box\varphi \supset \Box\Box\varphi$

$SemT:$	reflexive(R)
$SemB:$	symmetric(R)
$Sem4:$	transitive(R)

S5:	$AxT \wedge Ax4$	$\models F_n$
KTb:	$AxT \wedge Ax4$	$\models F_n$
S4:	$AxT \wedge Ax4$	$\models F_n$
KB4:	$AxB \wedge Ax4$	$\models F_n$
KT:	AxT	$\models F_n$
KB:	AxB	$\models F_n$
K4:	$Ax4$	$\models F_n$
K:		$\models F_n$

S5:	$SemT \wedge Sem4$	$\models F_n$
KTb:	$SemT \wedge Sem4$	$\models F_n$
S4:	$SemT \wedge Sem4$	$\models F_n$
KB4:	$SemB \wedge Sem4$	$\models F_n$
KT:	$SemT$	$\models F_n$
KB:	$SemB$	$\models F_n$
K4:	$Sem4$	$\models F_n$
K:		$\models F_n$

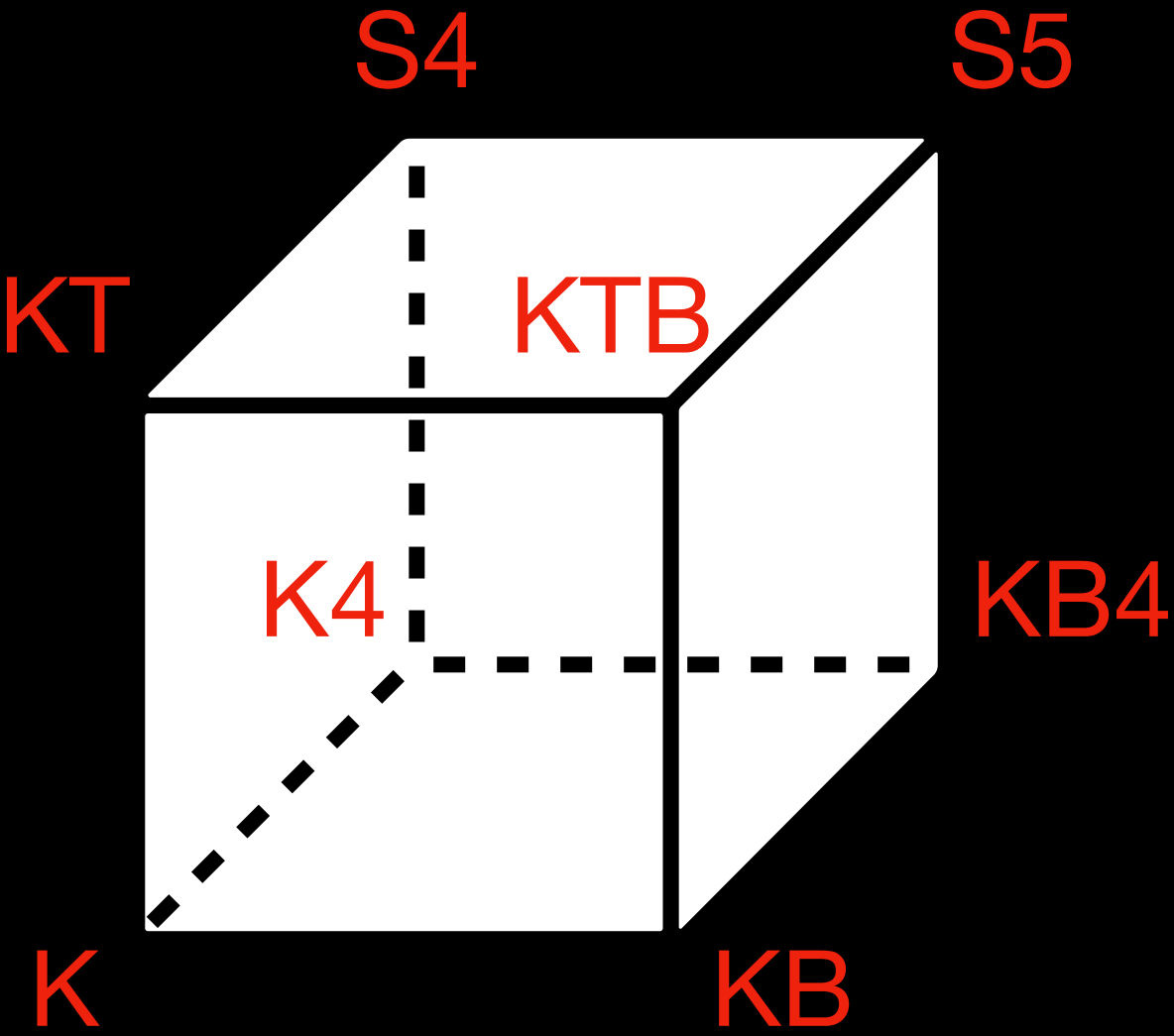
80 problems
with/without “apply simp”

80 problems
with/without “apply simp”

$$M \models F$$

$$N \models F$$

(schematic)
problem formulas



Problems	nitpick			sledgehammer	
	A	B	C	D	E
	counterex.	no counterex.	unknown	proof found	no proof
$M \models^m F$ $N \models^m F$	minimal shallow			minimal shallow	
$M \models^s F$ $N \models^s F$	maximal shallow			maximal shallow	
$M \models^d F$ $N \models^d F$	deep			deep	

Sledgehammer and Nitpick were run in default settings in Isabelle-2025-RC2

Evaluation

A MacBook Pro 16,1 (6-Core Intel Core i7 processor, 2,6 GHz, 6 Cores, 256 KB L2 Cache per Core, 12 MB L3 Cache, 16 GB Memory) and Isabelle 2025-1-RC1 in standard configuration was used.

		sledgehammer ¹⁴ (38 THM)									nitpick (42 CSA)				
		1s	2s	5s	10s	20s	40s	80s	Σ	%	2s	10s	40s	Σ	%
$M \models^m F$		9	9	12	19	24	34	37	144	45%	42	42	42	126	100%
$M \models^m F$	simp	18	36	37	38	38	38	38	243	91%	42	42	42	126	100%
$N \models^m F$		38	38	38	38	38	38	38	266	100%	42	42	42	126	100%
$N \models^m F$	simp	38	38	38	38	38	38	38	266	100%	42	42	42	126	100%
$M \models^s F$		0	0	0	23	23	24	29	99	37%	16	16	16	48	38%
$M \models^s F$	simp	7	26	26	28	28	28	28	181	68%	16	16	16	48	38%
$N \models^s F$		7	7	8	37	37	37	37	170	64%	42	42	42	126	100%
$N \models^s F$	simp	38	38	38	38	38	38	38	266	100%	42	42	42	126	100%
$M \models^d F$		7	12	21	24	26	30	30	150	56%	0	6	6	12	10%
$M \models^d F$	simp	0	7	13	26	28	28	28	130	49%	2	8	8	18	14%
$N \models^d F$		25	38	38	38	38	38	38	253	95%	6	24	24	54	43%
$N \models^d F$	simp	30	32	38	38	38	38	38	252	95%	2	8	8	18	14%

minimal shallow with modal axioms
minimal shallow with semantic axioms
maximal shallow with modal axioms
maximal shallow with semantic axioms
deep with modal axioms
deep with semantic axioms

Fig. 10. Performance results of the automated PML formula classification task conducted with sledgehammer and nitpick; for the problem set $M \models^m F$ a perfect classification is achieved (with or without the use of simp) already for a very short timeout of 2s (and also 1s), meaning that for each formula F in this mode either a proof is reported, if F is a theorem (THM), or a counterexample is presented, if F is a countersatisfiable (CSA). There were no conflicting results reported by sledgehammer and nitpick.

Results

- Winner(s) — perfect classification — “no counterex.”/“proof” for all valid statement and “counterex.” else

Evaluation

A MacBook Pro 16,1 (6-Core Intel Core i7 processor, 2,6 GHz, 6 Cores, 256 KB L2 Cache per Core, 12 MB L3 Cache, 16 GB Memory) and Isabelle 2025-1-RC1 in standard configuration was used.

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$M \models^d F$		7	12	21	24	26	30	30	150	56%	0	6	6	12	10%
$M \models^d F$	simp	0	7	13	26	28	28	28	130	49%	2	8	8	18	14%
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- minimal shallow with modal axioms
- minimal shallow with semantic axioms
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- maximal shallow with semantic axioms
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Results

- Winner — perfect classification — “no counterex.”/“proof” for all valid statement and “counterex.” else
- Second best

Evaluation

A MacBook Pro 16,1 (6-Core Intel Core i7 processor, 2,6 GHz, 6 Cores, 256 KB L2 Cache per Core, 12 MB L3 Cache, 16 GB Memory) and Isabelle 2025-1-RC1 in standard configuration was used.

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- Third best

Evaluation

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$N \models^m F$	simp	38	38	38	38	38	38	38	266	100%	42	42	42	126	100%
$M \models^s F$		0	0	0	23	23	24	29	99	37%	16	16	16	48	38%
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$M \models^d F$	simp	0	7	13	26	28	28	28	130	49%	2	8	8	18	14%
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Comments

- Remember that reasoning for **schematic formulas** was tested here
- Approach supports the **parallelization** of deep, minimal shallow and maximal shallow

**Does this approach scale for
Quantifiers? (Example: QBF)**

(new — extending results from CADE 2025)

QBF — Syntax and Semantics

The *syntax* of QBF with free variables is defined by the following grammar, where $a \in \mathcal{S}$ are propositional variable symbols declared in a (nonempty) signature $\mathcal{S}$:

$$\varphi, \psi := a \mid \neg\varphi \mid \varphi \vee \psi \mid \forall a.\varphi$$

Further logical connectives can be defined as usual, e.g.: $\varphi \supset \psi := \neg\varphi \vee \psi$, $\varphi \wedge \psi := \neg(\varphi \supset \neg\psi)$, $\exists a.\varphi := \neg\forall a.\neg\varphi$, $\top := a \supset a$ (for some $a \in \mathcal{S}$), and $\perp := \neg\top$.

A *semantics* for QBF determines the truth of a formula φ relative to an assignment function I , which assigns a truth value to each symbol $a \in \mathcal{S}$. Formally, we have:

$$\begin{aligned} I \models a & \text{ iff } I(a) \\ I \models \neg\varphi & \text{ iff not } I \models \varphi \\ I \models \varphi \vee \psi & \text{ iff } I \models \varphi \text{ or } I \models \psi \\ I \models \forall a.\varphi & \text{ iff for all } v \in \{T, F\} \text{ we have } I\langle a \leftarrow v \rangle \models \varphi \end{aligned}$$

The modified assignment function $I\langle a \leftarrow v \rangle$ in the last clause is defined such that it is identical to I except on a , for which it now maps to v .

Based on the above definitions, a notion of validity can be defined: A formula φ is valid iff $I \models \varphi$ for all assignment functions I .

QBF: Linked Embeddings

$$|\ " \forall^m x. \Phi x \equiv \forall x. \Phi x \"$$

$$|\ " \forall^s a. \varphi \equiv \lambda l. \forall v. \varphi (l \langle a \leftarrow v \rangle) \"$$

$$|\ " l \models^d \forall^d x. \varphi = (\forall v. l \langle x \leftarrow v \rangle \models^d \varphi) \"$$

shallow minimal ($\cdot^m$)

```

1 theory QBFinHOL_shallow_minimal imports QBFinHOL_preliminaries
2 begin
3 type_synonym  $\sigma = \text{bool}$ 
4 —<Surjective assignment functions are assumed at the meta-level>
5 consts IF:: $\mathcal{I}$  axiomatization where IF_surj: "surj IF"
6 —<Minimal shallow embedding (of QBF in HOL)>
7 definition VarM :: " $\mathcal{S} \Rightarrow \sigma$ " ("_m") where " $\forall^m v \equiv \text{IF } v$ "
8 definition NotM :: " $\sigma \Rightarrow \sigma$ " ("¬_m") where " $\neg^m \varphi \equiv \neg \varphi$ "
9 definition OrM :: " $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixr " $\vee^m$ " 930) where " $\varphi \vee^m \psi \equiv \varphi \vee \psi$ "
10 definition AllM :: " $(\mathcal{S} \Rightarrow \sigma) \Rightarrow \sigma$ " (binder " $\forall^m$ " 910) where " $\forall^m x. \Phi x \equiv \forall x. \Phi x$ "
11 —<Further logical connectives as derived definitions>
12 definition ImpM :: " $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixr " $\supset^m$ " 92) where " $\varphi \supset^m \psi \equiv (\neg^m \varphi) \vee^m \psi$ "
13 definition AndM :: " $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixr " $\wedge^m$ " 95) where " $\varphi \wedge^m \psi \equiv \neg^m (\varphi \supset^m \neg^m \psi)$ "
14 definition ExM :: " $(\mathcal{S} \Rightarrow \sigma) \Rightarrow \sigma$ " (binder " $\exists^m$ " 910) where " $\exists^m x. \Phi x \equiv \neg^m \forall^m x. \neg^m (\Phi x)$ "
15 consts p:: $\mathcal{S}$ 
16 definition TopM :: " $\sigma$ " ("⊤m") where "⊤m ≡ pm ⊃m pm"
17 definition BotM :: " $\sigma$ " ("⊥m") where "⊥m ≡ ¬m ⊤m"
18 —<Validity>
19 definition ValM :: " $\sigma \Rightarrow \text{bool}$ " ("⊨m" 9) where "⊨m  $\varphi \equiv \varphi$ "
20 —<Collection of definitions in a bag called DefS>
21 named_theorems DefM declare VarM_def[DefM] NotM_def[DefM] OrM_def[DefM] AllM_def[DefM]
22 ImpM_def[DefM] AndM_def[DefM] ExM_def[DefM] TopM_def[DefM] BotM_def[DefM] ValM_def[DefM]
23 end

```

```

1 theory QBFinHOL_deep imports QBFinHOL_preliminaries
2 begin
3 —<Deep embedding (of QBF in HOL)>
4 datatype  $\mathcal{F} = \text{VarD } \mathcal{S} ("^d") \mid \text{NotD } \mathcal{F} ("¬^d") \mid \text{OrD } \mathcal{F} \mathcal{F} (\text{infixr } "\vee^d" 925) \mid \text{AllD } \mathcal{S} \mathcal{F} (" \forall^d ")$ 
5 —<Further logical connectives as definitions>
6 definition ImpD (infixr " $\supset^d$ " 92) where " $\varphi \supset^d \psi \equiv (\neg^d \varphi) \vee^d \psi$ "
7 definition AndD (infixr " $\wedge^d$ " 95) where " $\varphi \wedge^d \psi \equiv \neg^d (\varphi \supset^d \neg^d \psi)$ "
8 definition ExD ("∃d _") where " $\exists^d a. \varphi \equiv \neg^d (\forall^d a. \neg^d \varphi)$ "
9 consts p ::  $\mathcal{S}$ 
10 definition TopD ("⊤d") where "⊤d ≡ pd ⊃d pd"
11 definition BotD ("⊥d") where "⊥d ≡ ¬d ⊤d"
12 —<Relative truth>
13 primrec RelativeTruthD ("⊨d" [100,100] 100) where
14   | "l ⊨d (ad)"       = l a
15   | "l ⊨d (¬d  $\varphi$ )"     = (¬ l ⊨d  $\varphi$ )
16   | "l ⊨d  $\varphi \vee^d \psi$ "  = (l ⊨d  $\varphi \vee$  l ⊨d  $\psi$ )
17   | "l ⊨d  $\forall^d x. \varphi$ "  = (∀ v. l ⟨x ← v⟩ ⊨d  $\varphi$ )
18 —<Validity>
19 definition ValD ("⊨d" 9) where "⊨d  $\varphi \equiv \forall l. l \models^d \varphi$ "
20 —<Collection of definitions in a bag called DefD>
21 named_theorems DefD
22 declare ImpD_def[DefD] AndD_def[DefD] ExD_def[DefD] TopD_def[DefD] BotD_def[DefD] ValD_def[DefD]
23 end

```

deep ($\cdot^d$)

```

1 theory QBFinHOL_shallow imports QBFinHOL_preliminaries
2 begin
3 type_synonym  $\sigma = \mathcal{I} \Rightarrow \text{bool}$ 
4 —<Maximal shallow embedding (of QBF in HOL)>
5 definition VarS :: " $\mathcal{S} \Rightarrow \sigma$ " ("_s") where " $a^s \equiv \lambda l. l a$ "
6 definition NotS :: " $\sigma \Rightarrow \sigma$ " ("¬s _") where " $\neg^s \varphi \equiv \lambda l. \neg (\varphi l)$ "
7 definition OrS :: " $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixr " $\vee^s$ " 930) where " $\varphi \vee^s \psi \equiv \lambda l. \varphi l \vee \psi l$ "
8 definition AllS :: " $(\mathcal{S} \Rightarrow \sigma) \Rightarrow \sigma$ " ("∀s _") where " $\forall^s a. \varphi \equiv \lambda l. \forall v. \varphi (l \langle a \leftarrow v \rangle)$ "
9 —<Further logical connectives as derived definitions>
10 definition ImpS :: " $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixr " $\supset^s$ " 92) where " $\varphi \supset^s \psi \equiv (\neg^s \varphi) \vee^s \psi$ "
11 definition AndS :: " $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixr " $\wedge^s$ " 95) where " $\varphi \wedge^s \psi \equiv \neg^s (\varphi \supset^s \neg^s \psi)$ "
12 definition ExS :: " $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " ("∃s _") where " $\exists^s a. \varphi \equiv \neg^s (\forall^s a. \neg^s \varphi)$ "
13 consts q ::  $\mathcal{S}$ 
14 definition TopS :: " $\sigma$ " ("⊤s") where "⊤s ≡ qs ⊃s qs"
15 definition BotS :: " $\sigma$ " ("⊥s") where "⊥s ≡ ¬s ⊤s"
16 —<Relative truth>
17 definition RelTruthS :: " $\mathcal{I} \Rightarrow \sigma \Rightarrow \text{bool}$ " (infix "⊨s" 10) where "l ⊨s  $\varphi \equiv \varphi l$ "
18 —<Validity>
19 definition ValS :: " $\sigma \Rightarrow \text{bool}$ " ("⊨s" 9) where "⊨s  $\varphi \equiv \forall l. l \models^s \varphi$ "
20 —<Collection of definitions in a bag called DefS>
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22 ImpS_def[DefS] AndS_def[DefS] ExS_def[DefS] TopS_def[DefS] BotS_def[DefS] RelTruthS_def[DefS] ValS_def[DefS]
23 end

```

shallow maximal ($\cdot^s$)

QBF: Linked Embeddings

shared preliminaries

```
1 theory QBFinHOL_preliminaries imports Main
2 begin
3 <Type declarations common for both the deep and shallow embedding>
4 type_synonym S = nat <Signature of variable symbols: we simply use the natural numbers for this>
5 type_synonym I = "S => bool" <Assignment functions>
6 <Modification of assignment functions>
7 definition IntMod :: "I => S => bool => I" ("<_<-_>") where "I(x<-y) ≡ λz. if z = x then y else I z"
8 <Some lemmata>
9 lemma L1 [simp]: "(I(y<-s)) x = I x" if "x ≠ y" using that IntMod_def by auto
10 lemma L2: "a ≠ c => (I(a<-b)<(c<-d)) = (I(c<-d)<(a<-b))" using IntMod_def by auto
11 lemma L3 [simp]: "(I(a<-b)) a = b" using IntMod_def by auto
12 lemma L4 [simp]: "(I(a<-b)<(a<-c)) = (I(a<-c))" using IntMod_def by auto
13 <Further settings> declare[[syntax_ambiguity_warning=false]] nitpick_params[user_axioms,expect=genuine]
14 end
```

shallow minimal ($\cdot^m$)

```
1 theory QBFinHOL_shallow_minimal imports QBFinHOL_preliminaries
2 begin
3 type_synonym σ = bool
4 <Surjective assignment functions are assumed at the meta-level>
5 consts IF::I axiomatization where IF_surj: "surj IF"
6 <Minimal shallow embedding (of QBF in HOL)>
7 definition VarM :: "S => σ" ("<_>^m") where "v^m ≡ IF v"
8 definition NotM :: "σ => σ" ("<_>^m") where "¬^m φ ≡ ¬ φ"
9 definition OrM :: "σ => σ => σ" (infix "<_>^m" 930) where "φ ∨^m ψ ≡ φ ∨ ψ"
10 definition AllM :: "(S => σ) => σ" (binder "∀^m" 910) where "∀^m x. φ x ≡ ∀ x. φ x"
11 <Further logical connectives as derived definitions>
12 definition ImpM :: "σ => σ => σ" (infix "<_>^m" 92) where "φ ⊃^m ψ ≡ (¬^m φ) ∨^m ψ"
13 definition AndM :: "σ => σ => σ" (infix "<_>^m" 95) where "φ ∧^m ψ ≡ ¬^m (φ ⊃^m ¬^m ψ)"
14 definition ExM :: "(S => σ) => σ" (binder "∃^m" 910) where "∃^m x. φ x ≡ ¬^m ∀^m x. ¬^m (φ x)"
15 consts p::S
16 definition TopM :: "σ" ("⊤^m") where "⊤^m ≡ p^m ⊃^m p^m"
17 definition BotM :: "σ" ("⊥^m") where "⊥^m ≡ ¬^m ⊤^m"
18 <Validity>
19 definition ValM :: "σ => bool" ("⊨^m" 9) where "⊨^m φ ≡ φ"
20 <Collection of definitions in a bag called DefS>
21 named_theorems DefM declare VarM_def[DefM] NotM_def[DefM] OrM_def[DefM] AllM_def[DefM]
22 ImpM_def[DefM] AndM_def[DefM] ExM_def[DefM] TopM_def[DefM] BotM_def[DefM] ValM_def[DefM]
23 end
```

```
1 theory QBFinHOL_deep imports QBFinHOL_preliminaries
2 begin
3 <Deep embedding (of QBF in HOL)>
4 datatype F = VarD S ("<_>^d") | NotD F ("¬^d") | OrD F F (infix "<_>^d" 925) | AllD S F ("∀^d")
5 <Further logical connectives as definitions>
6 definition ImpD (infix "<_>^d" 92) where "φ ⊃^d ψ ≡ (¬^d φ) ∨^d ψ"
7 definition AndD (infix "<_>^d" 95) where "φ ∧^d ψ ≡ ¬^d (φ ⊃^d ¬^d ψ)"
8 definition ExD ("∃^d") where "∃^d a. φ ≡ ¬^d (∀^d a. ¬^d φ)"
9 consts p :: S
10 definition TopD ("⊤^d") where "⊤^d ≡ p^d ⊃^d p^d"
11 definition BotD ("⊥^d") where "⊥^d ≡ ¬^d ⊤^d"
12 <Relative truth>
13 primrec RelativeTruthD ("⊨^d" [100,100] 100) where
14   "⊨^d (a^d) = I a"
15   "⊨^d (¬^d φ) = (¬ I ⊨^d φ)"
16   "⊨^d (φ ∨^d ψ) = (I ⊨^d φ ∨ I ⊨^d ψ)"
17   "⊨^d (∀^d x. φ) = (∀ v. I(x<-v) ⊨^d φ)"
18 <Validity>
19 definition ValD ("⊨^d" 9) where "⊨^d φ ≡ ∀ I. I ⊨^d φ"
20 <Collection of definitions in a bag called DefD>
21 named_theorems DefD
22 declare ImpD_def[DefD] AndD_def[DefD] ExD_def[DefD] TopD_def[DefD] BotD_def[DefD] ValD_def[DefD]
23 end
```

deep ($\cdot^d$)

```
1 theory QBFinHOL_shallow imports QBFinHOL_preliminaries
2 begin
3 type_synonym σ = "I => bool"
4 <Maximal shallow embedding (of QBF in HOL)>
5 definition VarS :: "S => σ" ("<_>^s") where "a^s ≡ λ I. I a"
6 definition NotS :: "σ => σ" ("<_>^s") where "¬^s φ ≡ λ I. ¬ (φ I)"
7 definition OrS :: "σ => σ => σ" (infix "<_>^s" 930) where "φ ∨^s ψ ≡ λ I. φ I ∨ ψ I"
8 definition AllS :: "(S => σ) => σ" ("∀^s") where "∀^s a. φ ≡ λ I. ∀ v. φ (I(a<-v))"
9 <Further logical connectives as derived definitions>
10 definition ImpS :: "σ => σ => σ" (infix "<_>^s" 92) where "φ ⊃^s ψ ≡ (¬^s φ) ∨^s ψ"
11 definition AndS :: "σ => σ => σ" (infix "<_>^s" 95) where "φ ∧^s ψ ≡ ¬^s (φ ⊃^s ¬^s ψ)"
12 definition ExS :: "(S => σ) => σ" ("∃^s") where "∃^s a. φ ≡ ¬^s (∀^s a. ¬^s φ)"
13 consts q :: S
14 definition TopS :: "σ" ("⊤^s") where "⊤^s ≡ q^s ⊃^s q^s"
15 definition BotS :: "σ" ("⊥^s") where "⊥^s ≡ ¬^s ⊤^s"
16 <Relative truth>
17 definition RelTruthS :: "I => σ => bool" (infix "⊨^s" 10) where "I ⊨^s φ ≡ φ I"
18 <Validity>
19 definition ValS :: "σ => bool" ("⊨^s" 9) where "⊨^s φ ≡ ∀ I. I ⊨^s φ"
20 <Collection of definitions in a bag called DefS>
21 named_theorems DefS declare VarS_def[DefS] NotS_def[DefS] OrS_def[DefS] AllS_def[DefS]
22 ImpS_def[DefS] AndS_def[DefS] ExS_def[DefS] TopS_def[DefS] BotS_def[DefS] RelTruthS_def[DefS] ValS_def[DefS]
23 end
```

shallow maximal ($\cdot^s$)

QBF — Faithfulness Automated

deep ($\cdot^d$)

```
1 theory QBFInHOL_deep imports QBFInHOL_preliminaries
2 begin
3   <Deep embedding (of QBF in HOL)>
4   datatype  $\mathcal{F}$  = VarD  $\mathcal{S}$  (" $\cdot^d$ ") | NotD  $\mathcal{F}$  (" $\neg^d$ ") | OrD  $\mathcal{F}$  (" $\vee^d$ ") | AndD  $\mathcal{F}$  (" $\wedge^d$ ")
5   <Further logical connectives as definitions>
6   definition ImpD (infix " $\supset^d$ " 92) where
7     " $\varphi \supset^d \psi \equiv (\neg^d \varphi) \vee^d \psi$ "
8   definition AndD (infix " $\wedge^d$ " 95) where
9     " $\varphi \wedge^d \psi \equiv \neg^d (\varphi \supset^d \neg^d \psi)$ "
10  definition ExD (" $\exists^d$ " 90) where
11    " $\exists^d a. \varphi \equiv \neg^d (\forall^d a. \neg^d \varphi)$ "
12  consts p ::  $\mathcal{S}$ 
13  definition TopD (" $\top^d$ ") where
14    " $\top^d \equiv p \supset^d p$ "
15  definition BotD (" $\bot^d$ ") where
16    " $\bot^d \equiv \neg^d \top^d$ "
17  <Relative truth>
18  primrec RelativeTruthD (" $\models^d$ " [100,100] 100) where
19    " $\models^d (a^d) \equiv \top^d$ "
20    " $\models^d (\neg^d \varphi) \equiv (\neg^d \models^d \varphi)$ "
21    " $\models^d (\varphi \vee^d \psi) \equiv (\models^d \varphi \vee \models^d \psi)$ "
22    " $\models^d (\varphi \wedge^d \psi) \equiv (\models^d \varphi \wedge \models^d \psi)$ "
23    " $\models^d (\exists^d x. \varphi) \equiv (\forall v. \models^d (\varphi))$ "
24  <Validity>
25  definition ValD (" $\models^d$ " 9) where
26    " $\models^d \varphi \equiv \forall v. \models^d \varphi$ "
27  <Collection of definitions in a bag called DefD>
28  named_theorems DefD
29  declare ImpD_def[DefD] AndD_def[DefD] ExD_def[DefD] TopD_def[DefD] BotD_def[DefD] ValD_def[DefD]
30 end
```

shallow maximal ($\cdot^s$)

```
1 theory QBFInHOL_shallow imports QBFInHOL_preliminaries
2 begin
3   type_synonym  $\sigma$  = " $\mathcal{I} \Rightarrow \text{bool}$ "
4   <Maximal shallow embedding (of QBF in HOL)>
5   definition VarS :: " $\mathcal{S} \Rightarrow \sigma$ " ("s") where
6     " $s \equiv \lambda l. \top$ "
7   definition NotS :: " $\sigma \Rightarrow \sigma$ " ("¬s") where
8     " $\neg^s \varphi \equiv \lambda l. \neg (\varphi l)$ "
9   definition OrS :: " $\sigma \Rightarrow \sigma$ " ("∨s") where
10    " $\varphi \vee^s \psi \equiv \lambda l. \varphi l \vee \psi l$ "
11  definition AndS :: " $\sigma \Rightarrow \sigma$ " ("∧s") where
12    " $\varphi \wedge^s \psi \equiv \lambda l. \forall v. \varphi (l \leftarrow v)$ "
13  <Further logical connectives as derived definitions>
14  definition ImpS :: " $\sigma \Rightarrow \sigma$ " (infix " $\supset^s$ " 92) where
15    " $\varphi \supset^s \psi \equiv (\neg^s \varphi) \vee^s \psi$ "
16  definition AndS :: " $\sigma \Rightarrow \sigma$ " (infix " $\wedge^s$ " 95) where
17    " $\varphi \wedge^s \psi \equiv \neg^s (\varphi \supset^s \neg^s \psi)$ "
18  definition ExS :: " $\sigma \Rightarrow \sigma$ " ("∃s") where
19    " $\exists^s a. \varphi \equiv \neg^s (\forall^s a. \neg^s \varphi)$ "
20  consts q ::  $\mathcal{S}$ 
21  definition TopS :: " $\sigma$ " ("⊤s") where
22    " $\top^s \equiv q \supset^s q$ "
23  definition BotS :: " $\sigma$ " ("⊥s") where
24    " $\bot^s \equiv \neg^s \top^s$ "
25  <Relative truth>
26  definition RelTruthS :: " $\mathcal{I} \Rightarrow \text{bool}$ " (infix " $\models^s$ " 10) where
27    " $\models^s \varphi \equiv \varphi \top^s$ "
28  <Validity>
29  definition ValS :: " $\sigma \Rightarrow \text{bool}$ " ("⊨s") 9) where
30    " $\models^s \varphi \equiv \forall l. \models^s \varphi$ "
31  <Collection of definitions in a bag called DefS>
32  named_theorems DefS declare VarS_def[DefS] NotS_def[DefS] OrS_def[DefS] AndS_def[DefS] ExS_def[DefS] TopS_def[DefS] BotS_def[DefS] RelTruthS_def[DefS] ValS_def[DefS]
33 end
```

shallow minimal ($\cdot^m$)

```
1 theory QBFInHOL_shallow_minimal imports QBFInHOL_preliminaries
2 begin
3   type_synonym  $\sigma$  = bool
4   <Surjective assignment functions are assumed at the meta-level>
5   consts IF ::  $\mathcal{I} \Rightarrow \text{bool}$  axiomatization where IF_surj: "surj IF"
6   <Minimal shallow embedding (of QBF in HOL)>
7   definition VarM :: " $\mathcal{S} \Rightarrow \sigma$ " ("m") where
8     " $m \equiv \text{IF}$ "
9   definition NotM :: " $\sigma \Rightarrow \sigma$ " ("¬m") where
10    " $\neg^m \varphi \equiv \neg \varphi$ "
11  definition OrM :: " $\sigma \Rightarrow \sigma$ " ("∨m") where
12    " $\varphi \vee^m \psi \equiv \varphi \vee \psi$ "
13  definition AndM :: " $\sigma \Rightarrow \sigma$ " ("∧m") where
14    " $\varphi \wedge^m \psi \equiv \forall x. \varphi \wedge \psi x$ "
15  <Further logical connectives as derived definitions>
16  definition ImpM :: " $\sigma \Rightarrow \sigma$ " (infix " $\supset^m$ " 92) where
17    " $\varphi \supset^m \psi \equiv (\neg^m \varphi) \vee^m \psi$ "
18  definition AndM :: " $\sigma \Rightarrow \sigma$ " (infix " $\wedge^m$ " 95) where
19    " $\varphi \wedge^m \psi \equiv \neg^m (\varphi \supset^m \neg^m \psi)$ "
20  definition ExM :: " $\sigma \Rightarrow \sigma$ " ("∃m") where
21    " $\exists^m a. \varphi \equiv \neg^m (\forall^m a. \neg^m \varphi)$ "
22  consts p ::  $\mathcal{S}$ 
23  definition TopM :: " $\sigma$ " ("⊤m") where
24    " $\top^m \equiv p \supset^m p$ "
25  definition BotM :: " $\sigma$ " ("⊥m") where
26    " $\bot^m \equiv \neg^m \top^m$ "
27  <Validity>
28  definition ValM :: " $\sigma \Rightarrow \text{bool}$ " ("⊨m") 9) where
29    " $\models^m \varphi \equiv \varphi$ "
30  <Collection of definitions in a bag called DefM>
31  named_theorems DefM declare VarM_def[DefM] NotM_def[DefM] OrM_def[DefM] AndM_def[DefM] ExM_def[DefM] TopM_def[DefM] BotM_def[DefM] ValM_def[DefM]
32 end
```

```
1 theory QBFInHOL_faithfulness
2 imports QBFInHOL_deep QBFInHOL_shallow_minimal QBFInHOL_shallow QBFInHOL_deep_subst lemma QBFInHOL_surj
3 begin
4   <Mappings: deep to maximal shallow and deep to minimal shallow>
5   fun DpToShS ("[") where
6     " $\llbracket x^d \rrbracket = x^s$ " | " $\llbracket \neg^d \varphi \rrbracket = \neg^s \llbracket \varphi \rrbracket$ " | " $\llbracket \varphi \vee^d \psi \rrbracket = \llbracket \varphi \rrbracket \vee^s \llbracket \psi \rrbracket$ " | " $\llbracket \forall^d v. \varphi \rrbracket = \forall^s v. \llbracket \varphi \rrbracket$ "
7   fun DpToShM ("[") where
8     " $\llbracket x^d \rrbracket = x^m$ " | " $\llbracket \neg^d \varphi \rrbracket = \neg^m \llbracket \varphi \rrbracket$ " | " $\llbracket \varphi \vee^d \psi \rrbracket = \llbracket \varphi \rrbracket \vee^m \llbracket \psi \rrbracket$ " | " $\llbracket \forall^d v. \varphi \rrbracket = (\forall^m x. (\llbracket v \leftarrow_r x^d \rrbracket (\varphi)))$ "
9   <Proving faithfulness between deep and maximal shallow>
10  theorem FaithfulSDlem: " $(\models^s \llbracket \varphi \rrbracket) \longleftrightarrow (\models^d \varphi)$ " by (induct  $\varphi$  arbitrary: I) (auto simp: DefS DefD)
11  theorem FaithfulSD: " $(\models^s \llbracket \varphi \rrbracket) \longleftrightarrow (\models^d \varphi)$ " by (simp add: FaithfulSDlem ValD_def ValS_def)
12  <Proving faithfulness between deep and minimal shallow>
13  theorem FaithfulMD: " $(\models^m \llbracket \varphi \rrbracket) \longleftrightarrow (\models^d \varphi)$ " by (induct  $\varphi$  rule: QInduct; simp add: DefD DefM; metis IF_surj surjE)
14  <Proving faithfulness between maximal and minimal shallow>
15  theorem FaithfulMS: " $(\models^m \llbracket \varphi \rrbracket) \longleftrightarrow (\models^s \llbracket \varphi \rrbracket)$ " using FaithfulSDlem FaithfulMD by auto
16 end
```


QBF: Lemmas for deep, e.g. substitution lemma

```
1 theory QBFInHOL_deep_subst_lemma imports QBFInHOL_deep
2 begin
3 primrec is_free (infix "free_in" 900) where
4   "x free_in (yd) = (y = x)" | "x free_in (¬dφ) = (x free_in φ)"
5   | "x free_in (φ ∨d ψ) = (x free_in φ ∨ x free_in ψ)" | "x free_in (∀dy. φ) = (x free_in φ ∧ x ≠ y)"
6 abbreviation is_not_free (infix "not_free_in" 900) where "x not_free_in φ ≡ ¬(x free_in φ)"
7 fun is_bound (infix "bound_in" 900) where
8   "x bound_in (yd) = False" | "x bound_in (¬dφ) = (x bound_in φ)"
9   | "x bound_in (φ ∨d ψ) = (x bound_in φ ∨ x bound_in ψ)" | "x bound_in (∀dy. φ) = (x = y ∨ (x bound_in φ))"
10 abbreviation is_not_bound (infix "not_bound_in" 900) where "x not_bound_in φ ≡ ¬(x bound_in φ)"
11 abbreviation occurs (infix "occurs_in" 900) where "x occurs_in φ ≡ x free_in φ ∨ x bound_in φ"
12 abbreviation not_in (infix "not_in" 900) where "x not_in φ ≡ x not_free_in φ ∧ x not_bound_in φ"
13 primrec fresh ("fresh'(_)"') where
14   "fresh xd = x+1" | "fresh ¬dφ = fresh φ" | "fresh (φ ∨d ψ) = max (fresh φ) (fresh ψ)" | "fresh ∀dx. φ = max (x+1) (fresh φ)"
15 lemma L5: "x bound_in φ ⇒ x < (fresh φ)" by (induct φ) auto
16 lemma L6: "x free_in φ ⇒ x < (fresh φ)" by (induct φ) auto
17 lemma L7: "(fresh φ) not_in φ" using L5 L6 by blast
18 lemma L8: "max (fresh φ) (fresh ψ) not_free_in φ" by (metis L6 L7 max.absorb3 max_def)
19 lemma L9: "max (fresh φ) (fresh ψ) not_bound_in φ" by (metis L5 L7 max.absorb3 max_def)
20 lemma L10: "max (fresh φ) (fresh ψ) not_free_in ψ" by (metis L8 max commute)
21 lemma L11: "max (fresh φ) (fresh ψ) not_bound_in ψ" by (metis L9 max commute)
22 lemma L12: "x not_free_in φ ⇒ (l(y ← s) ⊨d φ) = (l ⊨d φ)" by (induct φ arbitrary: l; simp) (metis L4 L2)
23 primrec Subst ("[_ ← _](_)"') where —<Substitution ignoring potential variable capture>
24   "[x ← τ](yd) = (if x = y then τ else (yd))" | "[x ← τ](¬dφ) = ¬d[x ← τ](φ)"
25   | "[x ← τ](φ ∨d ψ) = ([x ← τ](φ) ∨d [x ← τ](ψ))" | "[x ← τ](∀dy. φ) = (if x = y then (∀dy. φ) else (∀dy. [x ← τ](φ)))"
26 lemma L13[simp]: "size [x ← τ](φ) = size φ" by (induct φ; auto)
27 lemma L14[simp]: "[x ← xd](φ) = φ" by (induct φ; auto)
28 lemma L15: assumes "x ≠ a" shows "[a ← τ]([a ← xd](φ)) = [a ← xd](φ)" using assms by (induct φ) auto
29 lemma L16[simp]: assumes "a ≠ x" shows "a not_free_in ([a ← xd](φ))" using assms by (induct φ) auto
30 lemma SInduct: —<Induction over the size of a formula>
31   assumes "∧ x. P (xd)"
32   assumes "∧ φ. (∧ ψ. size ψ ≤ size φ ⇒ P ψ) ⇒ P (¬dφ)"
33   assumes "∧ φ ψ. (∧ χ. size χ ≤ size φ + size ψ ⇒ P χ) ⇒ P (φ ∨d ψ)"
34   assumes "∧ y φ. (∧ ψ. size ψ ≤ size φ ⇒ P ψ) ⇒ P (∀dy. φ)"
35   shows "P φ"
36   using assms proof (induct "size φ" arbitrary: φ rule: less_induct) case less thus ?case by (induct φ) auto qed
```

```
37 lemma QInduct: —<Stronger induction>
38   assumes "∧ x. P (xd)"
39   assumes "∧ φ. P φ ⇒ P (¬dφ)"
40   assumes "∧ φ ψ. P φ ⇒ P ψ ⇒ P (φ ∨d ψ)"
41   assumes "∧ y φ. (∧ ψ. size ψ ≤ size φ ⇒ P ψ) ⇒ P (∀dy. φ)"
42   shows "P φ" using assms by (induct φ rule: SInduct) auto
43 primrec SubstitutableForIn —<Substitutable for a variable>
44   ("_ is_subst_for _ in _" [999,1,999] 999) where "τ is_subst_for x in (yd) = True"
45   | "τ is_subst_for x in (¬dφ) = τ is_subst_for x in (φ)"
46   | "τ is_subst_for x in (φ ∨d ψ) = (τ is_subst_for x in (φ) ∧ τ is_subst_for x in (ψ))"
47   | "τ is_subst_for x in (∀dy. φ) = (y = x ∨ (x not_free_in φ ∨ y not_free_in τ) ∧ τ is_subst_for x in φ)"
48 lemma SubstitutionLemma [simp]: —<Substitution lemma>
49   assumes "τ is_subst_for x in φ" shows "(l ⊨d ([x ← τ](φ))) = (l(x ← (l ⊨d τ)) ⊨d φ)"
50   using assms by (induct φ arbitrary: l; auto simp: L12 L2)
51 fun ren_for_subst where —<Alphabetic variants for unrestricted substitution>
52   "ren_for_subst x τ (yd) = yd" | "ren_for_subst x τ (¬dφ) = (¬d(ren_for_subst x τ φ))"
53   | "ren_for_subst x τ (φ ∨d ψ) = (ren_for_subst x τ φ ∨d ren_for_subst x τ ψ)"
54   | "ren_for_subst x τ (∀dy. φ) = (if y free_in τ ∧ x free_in φ then
55     let f = fresh(φ ∨d τ); φ' = [y ← fd](φ) in (∀df. ren_for_subst x τ φ') else ∀dy. ren_for_subst x τ φ)"
56 lemma L17 [simp]: "size (ren_for_subst x τ φ) = size φ" by (induct φ arbitrary: τ x rule: QInduct; simp add: Let_def)
57 lemma L18: "α not_in φ ⇒ (αd) is_subst_for β in φ" by (induct φ) auto
58 lemma L19: "x free_in ψ ⇒ y ≠ x ⇒ x free_in [y ← τ](ψ)" by (induct ψ) auto
59 lemma L20 [simp]: "x free_in ren_for_subst x τ φ = (x free_in φ)"
60   by (induct φ rule: QInduct; simp add: Let_def) (metis L16 L6 L7 L19 max.absorb3 max_def_raw)
61 lemma L21: "(l ⊨d φ) = (l ⊨d (ren_for_subst x τ φ))" by (induct φ arbitrary: x τ l rule: QInduct;
62   simp add: Let_def; subst SubstitutionLemma; safe intro!: L18; insert L8 L9; simp; metis L8 L2 L12)
63 lemma L22: "τ is_subst_for x in (ren_for_subst x τ φ)" by (induct φ rule: QInduct; auto simp add: Let_def) (metis L10)
64 lemma L23: "(αd) is_subst_for α in φ" by (induct φ) auto
65 lemma L24: assumes "α not_free_in φ" shows "[α ← τ](φ) = φ" using assms by (induct φ) auto
66 lemma L25: "β not_free_in φ ⇒ (αd) is_subst_for β in φ" by (induct φ) auto
67 lemma L26 [simp]: "z bound_in [x ← yd](φ) = z bound_in φ" by (induct φ) auto
68 definition ren_subst ("[_ ← _](_)"') where "[x ← τ](φ) = [x ← τ](ren_for_subst x τ φ)" —<Renaming substitution>
69 lemma L27 [simp]: "(l ⊨d [x ← τ](φ)) = (l(x ← (l ⊨d τ)) ⊨d φ)" using L22 L21 unfolding ren_subst_def by force
70 lemma L28 [simp]: "size ([x ← τ](φ)) = size φ" by (induct φ rule: QInduct) (auto simp: ren_subst_def Let_def)
71 lemma L29: "l ⊨d (∀dx. φ) = (∀τ. l ⊨d ([x ← τ](φ)))" by (smt L27 RelativeTruthD.simps(4))
72 end
```


QBF: Lemmas (for deep), e.g. Substitution lemma

```
3 primrec is_free (infix "free'_in" 900) where  
4   "x free_in (yd) = (y = x)" | "x free_in (¬dφ) = (x free_in φ)"  
5   | "x free_in (φ ∨d ψ) = (x free_in φ ∨ x free_in ψ)" | "x free_in (∀dy. φ) = (x free_in φ ∧ x ≠ y)"  
6 abbreviation is_not_free (infix "not'_free'_in" 900) where "x not_free_in φ ≡ ¬(x free_in φ)"
```

```
23 primrec Subst ("[_←_]'"') where —<Substitution ignoring potential variable capture>  
24   "[x←τ](yd) = (if x = y then τ else (yd))" | "[x←τ](¬dφ) = ¬d[x←τ](φ)"  
25   | "[x←τ](φ ∨d ψ) = ([x←τ](φ) ∨d [x←τ](ψ))" | "[x←τ](∀dy. φ) = (if x = y then (∀dy. φ) else (∀dy. [x←τ](φ)))"
```

```
43 primrec SubstitutableForIn —<Substitutable for a variable>  
44   ("_ is'_subst'_for _ in _" [999,1,999] 999) where "τ is_subst_for x in (yd) = True"  
45   | "τ is_subst_for x in (¬dφ) = τ is_subst_for x in (φ)"  
46   | "τ is_subst_for x in (φ ∨d ψ) = (τ is_subst_for x in (φ) ∧ τ is_subst_for x in (ψ))"  
47   | "τ is_subst_for x in (∀dy. φ) = (y = x ∨ (x not_free_in φ ∨ y not_free_in τ) ∧ τ is_subst_for x in φ)"  
48 lemma SubstitutionLemma [simp]: —<Substitution lemma>  
49   assumes "τ is_subst_for x in φ" shows "(I ⊨d ([x←τ](φ))) = I(x←(I ⊨d τ)) ⊨d φ"  
50   using assms by (induction φ arbitrary: I; auto simp: L12 L2)
```


QBF: Discharging the surjectivity assumption (deep)

```
1 theory QBFinHOL_surj imports QBFinHOL_deep_subst_lemma
2 begin
3 theorem SurjectiveDeep:      " $(\forall I. I \models^d \varphi) \longleftrightarrow (\forall I. \text{surj } I \longrightarrow I \models^d \varphi)$ "
4   proof(safe) fix I
5     obtain a b where      " $a \neq b \wedge a \text{ not\_free\_in } \varphi \wedge b \text{ not\_free\_in } \varphi$ "      by (metis L10 L8 is_free.simps(1))
6     moreover define J where " $J \equiv \lambda x. \text{if } x \text{ free\_in } \varphi \text{ then } I \ x \text{ else } x = a$ "
7     ultimately have      " $\text{surj } J$ "      by (smt surj_def)
8     moreover assume      " $\forall I. \text{surj } I \longrightarrow I \models^d \varphi$ "
9     ultimately have      " $J \models^d \varphi$ "      by blast
10    moreover have      " $\forall x. x \text{ free\_in } \varphi \longrightarrow J \ x = I \ x$ "      unfolding J_def by force
11    ultimately show      " $I \models^d \varphi$ "      by (induct  $\varphi$  arbitrary: I J; simp add: DefD; smt IntMod_def)
12  qed(auto)
13 end
```


Playing with QBF in Isabelle/HOL
(new — extending results from CADE 2025)


```

1 theory QBFinHOL_experiments imports QBFinHOL_deep QBFinHOL_shallow QBFinHOL_shallow_minimal
2 begin
3 abbreviation "(x::S)  $\equiv$  1" abbreviation "(y::S)  $\equiv$  2" abbreviation "(z::S)  $\equiv$  3"
4 lemma " $\models^d \forall^d x. \exists^d y. (x^d \vee^d y^d)$ " unfolding DefD apply simp by auto
5 lemma " $\models^s \forall^s x. \exists^s y. (x^s \vee^s y^s)$ " unfolding DefS apply simp by auto
6 lemma " $\models^m \forall^m x. \exists^m y. (x^m \vee^m y^m)$ " unfolding DefM apply simp by (metis IF_surj surjE)
7
8 lemma " $\models^d \forall^d x. \exists^d y. (x^d \wedge^d y^d)$ " unfolding DefD apply simp nitpick oops —<Counterexample>
9 lemma " $\models^s \forall^s x. \exists^s y. (x^s \wedge^s y^s)$ " unfolding DefS apply simp nitpick oops —<Counterexample>
10 lemma " $\models^m \forall^m x. \exists^m y. (x^m \wedge^m y^m)$ " unfolding DefM apply simp nitpick oops —<Counterexample>
11
12 lemma " $\models^d \forall^d x. \forall^d y. \exists^d z. (((z^d \supset^d ((x^d \supset^d \neg^d y^d) \wedge^d (y^d \supset^d \neg^d x^d))) \wedge^d (((x^d \supset^d \neg^d y^d) \wedge^d (y^d \supset^d \neg^d x^d)) \supset^d z^d)))$ "
13 unfolding DefD apply simp by auto
14 lemma " $\models^s \forall^s x. \forall^s y. \exists^s z. (((z^s \supset^s ((x^s \supset^s \neg^s y^s) \wedge^s (y^s \supset^s \neg^s x^s))) \wedge^s (((x^s \supset^s \neg^s y^s) \wedge^s (y^s \supset^s \neg^s x^s)) \supset^s z^s)))$ "
15 unfolding DefS apply simp by auto
16 lemma " $\models^m \forall^m x. \forall^m y. \exists^m z. (((z^m \supset^m ((x^m \supset^m \neg^m y^m) \wedge^m (y^m \supset^m \neg^m x^m))) \wedge^m (((x^m \supset^m \neg^m y^m) \wedge^m (y^m \supset^m \neg^m x^m)) \supset^m z^m)))$ "
17 unfolding DefM apply simp by (metis IF_surj surjE)
18 end

```


Fig. 22. A valid, moderate size QBF formula, termed QBF_{mediumD}, encoded in the deep embedding using ASCII prefix notation only.

Fig. 23. The formula QBFmediumD from Figure 22 as depicted by Isabelle/HOL after unfolding of definitions and simplification.


```

1 theory QBFinHOL_experiments_medium imports QBFinHOL_deep QBFinHOL_shallow QBFinHOL_shallow_minimal
2 begin
3   abbreviation "EquivD  $\varphi \psi \equiv$  (AndD (ImpD  $\varphi \psi$ ) (ImpD  $\psi \varphi$ ))"
4   lemma QBFmediumD: "
5      $\models^d$  (AIID 30 (ExD 29 (AIID 28 (ExD 27 (AIID 26 (ExD 25 (AIID 24 (ExD 23 (AIID 22 (ExD 21 (AIID 20 (ExD 19 (AIID 18 (ExD 17
6     (AIID 16 (ExD 15 (AIID 14 (ExD 13 (AIID 12 (ExD 11 (AIID 10 (ExD 9 (AIID 8 (ExD 7 (AIID 6 (ExD 5 (AIID 4 (ExD 3 (AIID 2 (ExD 1
7     (AndD (EquivD (AndD (AndD (EquivD (OrD (EquivD (VarD 24) (VarD 21)) (EquivD (VarD 1) (VarD 3))) (OrD (OrD (VarD 21)
8     (NotD (VarD 16))) (OrD (VarD 16) (VarD 20))) (ImpD (EquivD (EquivD (VarD 17) (NotD (VarD 15))) (EquivD (VarD 24)
9     (NotD (VarD 17)))) (ImpD (AndD (VarD 20) (VarD 19)) (EquivD (VarD 14) (VarD 1)))) (OrD (ImpD (AndD (ImpD (NotD (VarD 15))
10    (VarD 2)) (EquivD (VarD 24) (VarD 22))) (OrD (AndD (VarD 8) (VarD 7)) (EquivD (VarD 14) (NotD (VarD 14)))) (ImpD (EquivD
11    (ImpD (NotD (VarD 28)) (VarD 25)) (EquivD (VarD 19) (VarD 10))) (AndD (AndD (VarD 16) (NotD (VarD 25))) (EquivD (VarD 21)
12    (NotD (VarD 13)))) (AndD (AndD (EquivD (EquivD (OrD (VarD 10) (VarD 23)) (OrD (NotD (VarD 3)) (VarD 22))) (EquivD (OrD
13    (VarD 12) (NotD (VarD 21))) (AndD (VarD 4) (VarD 2))) (OrD (OrD (ImpD (VarD 23) (VarD 27)) (EquivD (VarD 26) (VarD 9)))
14    (AndD (ImpD (VarD 18) (NotD (VarD 8))) (AndD (VarD 19) (VarD 23)))) (EquivD (OrD (ImpD (ImpD (VarD 28) (VarD 2)) (EquivD
15    (NotD (VarD 6)) (VarD 6))) (ImpD (EquivD (VarD 26) (VarD 22)) (ImpD (NotD (VarD 25)) (VarD 30))) (EquivD (EquivD (EquivD
16    (VarD 28) (VarD 12)) (OrD (VarD 17) (NotD (VarD 29))) (EquivD (AndD (NotD (VarD 28)) (NotD (VarD 8))) (OrD (VarD 17)
17    (VarD 10)))) (EquivD (OrD (ImpD (EquivD (OrD (EquivD (VarD 5) (VarD 6)) (ImpD (VarD 21) (VarD 16))) (EquivD (OrD
18    (VarD 25) (VarD 12)) (ImpD (NotD (VarD 13)) (VarD 15))) (ImpD (EquivD (AndD (VarD 25) (NotD (VarD 4))) (EquivD (NotD
19    (VarD 26)) (VarD 26))) (ImpD (AndD (VarD 13) (NotD (VarD 11))) (ImpD (NotD (VarD 13)) (NotD (VarD 25)))) (EquivD (AndD
20    (EquivD (AndD (VarD 12) (VarD 18)) (ImpD (VarD 4) (VarD 14))) (ImpD (OrD (VarD 12) (NotD (VarD 14))) (ImpD (VarD 3) (VarD
21    20))) (AndD (ImpD (EquivD (VarD 27) (VarD 7)) (EquivD (NotD (VarD 2)) (VarD 1))) (AndD (OrD (NotD (VarD 10)) (VarD 14))
22    (EquivD (NotD (VarD 11)) (NotD (VarD 9)))) (ImpD (EquivD (AndD (EquivD (OrD (VarD 5) (VarD 7)) (EquivD (VarD 14)
23    (VarD 10))) (OrD (EquivD (NotD (VarD 15)) (VarD 9)) (EquivD (VarD 11) (VarD 3))) (EquivD (AndD (AndD (VarD 27) (NotD
24    (VarD 30))) (ImpD (NotD (VarD 2)) (NotD (VarD 30))) (ImpD (EquivD (NotD (VarD 4)) (VarD 13)) (OrD (VarD 24) (VarD 7))))
25    (EquivD (AndD (EquivD (OrD (VarD 2) (VarD 19)) (EquivD (VarD 24) (VarD 3))) (ImpD (OrD (VarD 3) (VarD 8)) (ImpD (VarD 6)
26    (VarD 28))) (AndD (AndD (OrD (NotD (VarD 30)) (VarD 8)) (EquivD (VarD 27) (VarD 27))) (ImpD (EquivD (VarD 14) (VarD 18))
27    (OrD (VarD 22) (NotD (VarD 26))))))))))))))))))))))))))))))))))))))))"
28   unfolding DefD apply simp sledgehammer[z3,timeout=4] oops —<Proof by z3 in 4s>
29 end

```

Fig. 22. A valid, moderate size QBF formula, termed QBFmediumD, encoded in the deep embedding using ASCII prefix notation only.

```

; smt.random_seed=1 smt.refine_ini_axioms=false -T:30 -smt2
(set-option :produce-proofs true)
(set-logic AUFLIA)
(assert (! (not (forall ((?v0 Bool)) (exists ((?v1 Bool)) (forall ((?v2 Bool)) (exists ((?v3 Bool)) (forall ((?v4 Bool)) (exists ((?v5 Bool)) (forall ((?v6 Bool))
(exists ((?v7 Bool)) (forall ((?v8 Bool)) (exists ((?v9 Bool)) (forall ((?v10 Bool)) (exists ((?v11 Bool)) (forall ((?v12 Bool)) (exists ((?v13 Bool)) (forall ((?
v14 Bool)) (exists ((?v15 Bool)) (forall ((?v16 Bool)) (exists ((?v17 Bool)) (forall ((?v18 Bool)) (exists ((?v19 Bool)) (forall ((?v20 Bool)) (exists ((?v21
Bool)) (forall ((?v22 Bool)) (exists ((?v23 Bool)) (forall ((?v24 Bool)) (exists ((?v25 Bool)) (forall ((?v26 Bool)) (exists ((?v27 Bool)) (forall ((?v28 Bool))
(exists ((?v29 Bool)) (and (or (and (or (and ?v6 (not ?v9)) (and ?v9 (not ?v6))) (or (and ?v29 (not ?v27)) (and ?v27 (not ?v29)))) (or (and ?v11 (and ?
v10 (and (or (and ?v13 ?v15) (or (and (not ?v15) (not ?v13)) (and (=> ?v6 (not ?v13)) (or ?v13 ?v6)))) (and (or (and ?v6 ?v13) (or (and (not ?v13)
(not ?v6)) (and (=> ?v13 (not ?v15)) (or ?v15 ?v13)))) (or (and ?v16 (not ?v29)) (and ?v29 (not ?v16)))))) (or (and (or ?v15 ?v28) (and (=> ?v6 ?v8)
(and (=> ?v8 ?v6) (and (=> ?v22 (not ?v23)) (and (or (and (not ?v2) (not ?v5)) (and (=> ?v11 ?v20) (=> ?v20 ?v11))) (and (or (and ?v11 (not ?v20)) (or
(and ?v20 (not ?v11)) (or ?v2 ?v5))) (=> ?v14 (or ?v5 (or (and ?v9 ?v17) (and (not ?v17) (not ?v9)))))))) (and (or (and ?v27 (and (or ?v20 ?v7) (not ?
v8))) (or (and (=> ?v27 ?v8) (and (not ?v20) (not ?v7))) (and (or (and (not ?v18) ?v9) (and ?v26 ?v28)) (=> (and ?v28 ?v26) (or ?v18 (not ?v9)))))) (and
(or (and (or ?v18 (not ?v9)) (=> ?v26 (not ?v28))) (or (and ?v28 (and ?v26 (and (not ?v18) ?v9))) (and (=> ?v27 (or (and (not ?v20) (not ?v7)) ?v8)) (or
(and ?v27 (not ?v8)) (or ?v20 ?v7)))) (and (=> ?v7 (or ?v3 (or (and (=> ?v4 ?v21) (=> ?v21 ?v4)) (and ?v11 (=> ?v12 (not ?v22)))))) (and (or (and
(=> ?v2 ?v28) (and (=> ?v4 ?v8) (and (=> ?v8 ?v4) (and (not ?v5) (not ?v0)))) (and (or (and (=> ?v2 ?v18) (and (=> ?v18 ?v2) (and (not ?v13) ?v1)))
(or (and (or ?v13 (not ?v1)) (or (and ?v2 (not ?v18)) (and ?v18 (not ?v2))) (and (or ?v2 (or ?v22 (or ?v13 ?v20))) (or (and (not ?v13) (not ?v20)) (and
(not ?v2) (not ?v22)))) (or (and (not ?v2) (and (not ?v22) (and (not ?v13) (not ?v20))) (or (and (or ?v13 ?v20) (or ?v2 ?v22)) (and (or (and ?v2 (not ?
v18)) (or (and ?v18 (not ?v2)) (or ?v13 (not ?v1))) (or (and (not ?v13) ?v1) (and (=> ?v2 ?v18) (=> ?v18 ?v2)))) (or (and (or (and ?v2 (not ?v18)) (or
(and ?v18 (not ?v2)) (or ?v13 (not ?v1))) (and (or (and (not ?v13) ?v1) (and (=> ?v2 ?v18) (=> ?v18 ?v2))) (or (and (not ?v2) (and (not ?v22) (and
(not ?v13) (not ?v20))) (and (or ?v13 ?v20) (or ?v2 ?v22)))) (or (and (or ?v2 (or ?v22 (or ?v13 ?v20))) (and (or (and (not ?v13) (not ?v20)) (and (not ?
v2) (not ?v22)) (or (and (=> ?v2 ?v18) (and (=> ?v18 ?v2) (and (not ?v13) ?v1))) (and (or ?v13 (not ?v1)) (or (and ?v2 (not ?v18)) (and ?v18 (not ?
v2)))))) (or (and ?v2 (not ?v28)) (or (and ?v4 (not ?v8)) (or (and ?v8 (not ?v4)) (or ?v5 ?v0)))))) (and (or (and (=> ?v27 (or (and (not ?v20) (not ?
v7)) ?v8)) (and (or (and ?v27 (not ?v8)) (or ?v20 ?v7)) (or (and (or ?v18 (not ?v9)) (=> ?v26 (not ?v28))) (and ?v28 (and ?v26 (and (not ?v18) ?v9)))) (and
(or (and (or (and (not ?v18) ?v9) (and ?v26 ?v28)) (and (=> (and ?v26 ?v28) (or ?v18 (not ?v9))) (or (and ?v27 (and (or ?v20 ?v7) (not ?v8))) (and
(=> ?v27 ?v8) (and (not ?v20) (not ?v7)))) (or (and ?v7 (and (not ?v3) (and (or (and ?v4 (not ?v21)) (and ?v21 (not ?v4))) (=> ?v11 (and ?v12 ?
v22)))) (or (and (or (and ?v2 (not ?v28)) (or (and ?v4 (not ?v8)) (or (and ?v8 (not ?v4)) (or ?v5 ?v0))) (or (and (or (and ?v2 (not ?v18)) (or (and ?v18
(not ?v2)) (or ?v13 (not ?v1))) (and (or (and (not ?v13) ?v1) (and (=> ?v2 ?v18) (=> ?v18 ?v2))) (or (and (not ?v2) (and (not ?v22) (and (not ?v13)
(not ?v20))) (and (or ?v13 ?v20) (or ?v2 ?v22)))) (and (or ?v2 (or ?v22 (or ?v13 ?v20))) (and (or (and (not ?v13) (not ?v20)) (and (not ?v2) (not ?
v22)) (or (and (=> ?v2 ?v18) (and (=> ?v18 ?v2) (and (not ?v13) ?v1))) (and (or ?v13 (not ?v1)) (or (and ?v2 (not ?v18)) (and ?v18 (not ?v2)))))) (or
(= and (or (and (=> ?v2 ?v18) (and (=> ?v18 ?v2) (and (not ?v13) ?v1))) (or (and (or ?v13 (not ?v1)) (or (and ?v2 (not ?v18)) (and ?v18 (not ?v2)))) (and
(or ?v2 (or ?v22 (or ?v13 ?v20))) (or (and (not ?v13) (not ?v20)) (and (not ?v2) (not ?v22)))) (and (or (and (not ?v2) (and (not ?v22) (and (not ?v13)
(not ?v20))) (or (and (or ?v13 ?v20) (or ?v2 ?v22)) (and (or (and ?v2 (not ?v18)) (or (and ?v18 (not ?v2)) (or ?v13 (not ?v1))) (or (and (not ?v13) ?v1)
(= and (=> ?v2 ?v18) (=> ?v18 ?v2)))) (and (=> ?v2 ?v28) (and (=> ?v4 ?v8) (and (=> ?v8 ?v4) (and (not ?v5) (not ?v0)))) (and (or (and (=> ?v6 ?v9)
(=> ?v9 ?v6)) (and (=> ?v29 ?v27) (=> ?v27 ?v29))) (and (=> (and ?v11 ?v10) (or (and (=> ?v13 (not ?v15)) (and (or ?v15 ?v13) (or (and ?v6 ?v13)
(= and (not ?v13) (not ?v6)))) (or (and (=> ?v6 (not ?v13)) (and (or ?v13 ?v6) (or (and ?v13 ?v15) (and (not ?v15) (not ?v13)))) (and (=> ?v16 ?v29)
(=> ?v29 ?v16)))) (or (and (not ?v15) (not ?v28)) (or (and ?v6 (not ?v8)) (or (and ?v8 (not ?v6)) (or (and ?v22 ?v23) (or (and (or ?v2 ?v5) (or (and ?
v11 (not ?v20)) (and ?v20 (not ?v11))) (or (and (=> ?v11 ?v20) (and (=> ?v20 ?v11) (and (not ?v2) (not ?v5)))) (and ?v14 (and (not ?v5) (and (=> ?v9
(not ?v17)) (or ?v17 ?v9)))))) (or (and (or (and (not ?v25) (not ?v23)) (and (=> ?v16 ?v20) (=> ?v20 ?v16))) (and (or (and ?v16 (not ?v20)) (or
(= and ?v20 (not ?v16)) (or ?v25 ?v23))) (and (or (and (or ?v15 ?v21) (=> ?v21 (not ?v15))) (and (=> ?v19 ?v27) (=> ?v27 ?v19))) (or (and ?v3 (and
(not ?v0) (and (or ?v28 (not ?v0)) (and (or ?v26 ?v17) (and (=> ?v17 (not ?v26)) (and (not ?v6) (not ?v23)))) (and (or (and (not ?v26) (not ?v17)) (or
(= and ?v17 ?v26) (or ?v6 ?v23))) (=> ?v3 (or ?v0 (and (not ?v28) ?v0)))) (or (and (=> ?v3 (or ?v0 (or (and (not ?v28) ?v0) (or (and (not ?v26) (not ?
v17)) (or (and ?v17 ?v26) (or ?v6 ?v23)))) (and (or (and (or ?v26 ?v17) (and (=> ?v17 (not ?v26)) (and (not ?v6) (not ?v23))) (and ?v3 (and (not ?v0)
(or ?v28 (not ?v0)))) (or (and (or ?v25 ?v23) (or (and ?v16 (not ?v20)) (and ?v20 (not ?v16))) (or (and (=> ?v16 ?v20) (and (=> ?v20 ?v16) (and
(not ?v25) (not ?v23))) (and (or (and (not ?v15) (not ?v21)) (and ?v21 ?v15)) (or (and ?v19 (not ?v27)) (and ?v27 (not ?v19)))) (and (or (and (or ?
v28 ?v11) (or (and ?v6 (not ?v27)) (and ?v27 (not ?v6))) (or (and (=> ?v6 ?v27) (and (=> ?v27 ?v6) (and (not ?v28) (not ?v11))) (or (and ?v24 (and
(or ?v27 ?v22) (not ?v2))) (and (=> ?v0 ?v22) (or (and ?v16 (not ?v12)) (or (and ?v12 (not ?v16)) (or ?v8 (not ?v4)))) (or (and ?v0 (not ?v22)) (or
(= and (=> ?v16 ?v12) (and (=> ?v12 ?v16) (and (not ?v8) ?v4))) (and (or (and (not ?v28) (not ?v11)) (and (=> ?v6 ?v27) (=> ?v27 ?v6))) (and (or (and ?
v6 (not ?v27)) (or (and ?v27 (not ?v6)) (or ?v28 ?v11)) (=> ?v24 (or (and (not ?v27) (not ?v22)) ?v2)))))))).named
conjecture0))
(check-sat)
(get-proof)

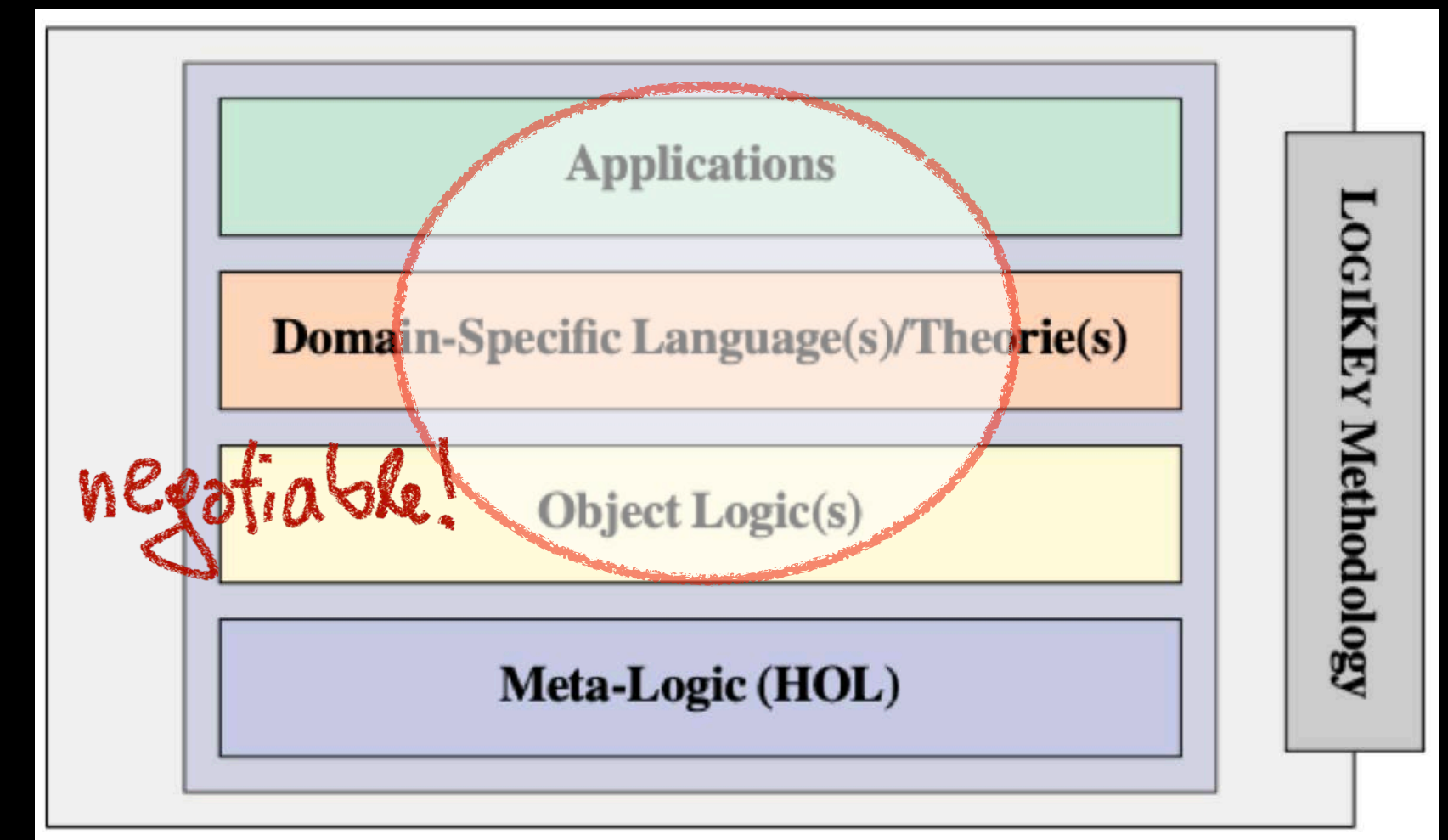
```

Screenshot

Fig. 24. Proof input file for z3 as generated by sledgehammer (in overlord mode) for the simplified formula QBFmediumD from Figure 22.

Discussion:

Virtues of approach



- supports the exploration of (expressive) **logics & logic combinations**
- supports the exploration & verification of **meta-logical properties**
- enables the **simultaneous use** of deep & shallow embeddings
- enables **integration of generic and specialist tools** (opportunities for further work)
- supports **uniform logic education** (with single working environment for students)
- has **innovative applications** (philosophy, law & ethics, foundations of maths, ...)
- creates **uniform proofs** at the meta-level (after unfolding)
- ...

Reading:

The presented material significantly extends the work covered in:

- Christoph Benz Müller: Faithful Logic Embeddings in HOL - Deep and Shallow. CADE 2025: 280-301. Doi: 10.1007/978-3-031-99984-0_16 (Preprint: [arXiv:2502.19311](https://arxiv.org/abs/2502.19311))
- Christoph Benz Müller: Faithful Logic Embeddings in HOL - Deep and Shallow (Isabelle/HOL dataset). Arch. Formal Proofs 2025 (2025). <https://www.isa-afp.org/entries/FaithfulPMLinHOL.html>

Further Reading

(preprints: <http://christoph-benzmueller.de/publications.html>)

HOL (see also the papers of Church, Henkin, Andrews and Muskens in JSL as mentioned before)

- Benzmüller, Brown, Kohlhase (2004), Higher-Order Semantics and Extensionality, J. Symb. Log., <http://doi.org/10.2178/jsl/1102022211>
- Benzmüller & Miller (2014), Automation of Higher-Order Logic, Handbook of the History of Logic, <http://doi.org/10.1016/B978-0-444-51624-4.50005-8>
- Benzmüller & Andrews (2019), Church's Type Theory, SEP, <https://plato.stanford.edu/entries/type-theory-church/>

Leo Provers

- Steen & Benzmüller (2021), Extensional Higher-Order Paramodulation in Leo-III, J. Autom. Reason., <http://doi.org/10.1007/s10817-021-09588-x>
- Benzmüller, Sultana, Paulson (2015), Thhe Higher-Order Prover Leo-II, J. Autom. Reason., <http://doi.org/10.1007/s10817-015-9348-y>
- Benzmüller & Kohlhase (1998), LEO – A Higher-Order Theorem Prover, CADE, <http://doi.org/10.1007/BFb0054256>

LogiKEy & Universal (Meta-)Logical Reasoning

- Benzmüller (2019), **Universal (meta-)logical reasoning: Recent successes**. Sci. Comp. Progr. (<http://doi.org/10.1016/j.scico.2018.10.008>)
- __, Parent & van der Torre (2020), **Designing normative theories for ethical and legal reasoning: LogiKEy framework, methodology, and tool support**. Artificial Intelligence. (<https://doi.org/10.1016/j.artint.2020.103348>)
- Benzmüller et.al. (2020), **LogiKEy Workbench: Deontic Logics, Logic Combinations and Expressive Ethical and Legal Reasoning (Isabelle/HOL Dataset)**. Data in Brief (<https://doi.org/10.1016/j.dib.2020.106409>)
- Bibel (2022), **Computer Kreiert Wissenschaft**. Informatik Spektrum. (<https://doi.org/10.1007/s00287-022-01456-1>)

Reading (cont'd)

(preprints: <http://christoph-benzmueller.de/publications.html>)

Foundational Studies in Metaphysics

- Kirchner, Benz Müller & Zalta (2019), **Computer Science and Metaphysics: A Cross-Fertilization**. Open Philosophy (<http://doi.org/10.1515/opphil-2019-0015>)
- __ (2020), **Mechanizing Principia Logico-Metaphysica in Functional Type Theory**. Review of Symbolic Logic (<https://doi.org/10.1017/S1755020319000297>)
- Kirchner (2021). **Embedding of Abstract Object Theory in Isabelle/HOL**. Full sources. (<https://github.com/ekpyron/AOT/tree/dissertation>)
- __ (2022), **Computer-Verified Foundations of Metaphysics and an Ontology of Natural Numbers in Isabelle/HOL**. PhD thesis, Dep. of Maths and CS, FU Berlin.

Foundational Studies in Ethics & Law

- Fuenmayor & Benz Müller (2018), **Formalisation and Evaluation of Alan Gewirth's Proof for the Principle of Generic Consistency in Isabelle/HOL**. (<https://www.isa-afp.org/entries/GewirthPGCProof.html>)
- __ (2019), **Harnessing Higher-Order (Meta-)Logic to Represent and Reason with Complex Ethical Theories**. PRICAI 2019 (https://doi.org/10.1007/978-3-030-29908-8_34)
- Benz Müller & Fuenmayor (2021), **Value-oriented Legal Argumentation in Isabelle/HOL**. ITP 2021 (<https://doi.org/10.4230/LIPIcs.ITP.2021.7>)
- __, Fuenmayor & Lomfeld (2022), **Modelling Value-oriented Legal Reasoning in LogiKey**. (<https://arxiv.org/abs/2006.12789>)

Studies on the Ontological Argument

- Benz Müller & Woltzenlogel Paleo (2014), **Automating Gödel's Ontological Proof of God's Existence with Higher-order Automated Theorem Provers**. ECAI 2014 (<http://doi.org/10.3233/978-1-61499-419-0-93>)
- __ (2016), **The Inconsistency in Gödel's Ontological Argument: A Success Story for AI in Metaphysics**. IJCAI 2016 (<http://www.ijcai.org/Proceedings/16/Papers/137.pdf>)

Reading (cont'd)

(preprints: <http://christoph-benzmueller.de/publications.html>)

Studies on the Ontological Argument (cont'd)

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